RENNER MASTERCLASS & CONFERENCE 2025 • BEIJING & TOYOHASHI

LOUIS RENNER

The Heart of the Piano: Over 140 Years of Tradition, Unified Manufacturing in Meuselwitz & Precision Hammerheads

Speaker: **Clemens Kunert**

Event: **Beijing & Toyohashi Conference 2025**

Headquarters: **Louis Renner GmbH • Meuselwitz, Germany**

OVERVIEW • OUTLINE

Structure and Chapters at a Glance

From our Meuselwitz headquarters and Wurzen felt manufacturing to hammerhead production, the Yamaha Kit, and voicing

01 02 03 04 05

CHAPTER 1 • Slides 3–4

History & Meuselwitz

Over 140 years of action and hammerhead tradition since 1882, plus construction insights and a drone flight through our unified Meuselwitz plant.

OPEN CHAPTER

CHAPTER 2 • Slides 5–10

Wurzen Felt Production

Filzfabrik Wurzen since 1783, premium wool blend, carding, defined fulling density, and the impact of leaving the felt sheets unsanded.

OPEN CHAPTER

CHAPTER 3 • Slides 11–18

Hammerhead Production

Felt storage, Weickert Special SEM analysis, cutting, hammer mouldings & hammer weight, pressing and gluing, CNC contour milling and precision surface calibration, plus riveting and boring.

OPEN CHAPTER

CHAPTER 4 • Slides 19–24

Yamaha Kit (C1–C3)

Ready-to-mount C1–C3 upgrade in 4–5 hours on site, time-lapse installation, pianist feedback & acoustic analysis (H_1–H_{12} to C_8).

OPEN CHAPTER

CHAPTER 5 • Slides 25–33

Voicing & Open Q&A

A_1 overtone buildup, complex spring, voicing model, crown and string plane, fine voicing, lacquer & ironing, plus Academy & open Q&A.

OPEN CHAPTER

PART 1: INTRODUCTION & HISTORY • COMPANY TIMELINE

140+ Years of Tradition & Perfection (1882 – Today)

From a Stuttgart artisan workshop to the global benchmark for piano actions & hammerheads

1882 • 1 of 9

Founder: Louis Renner (1882)

1882

Milestone 1 of 9

1882: Founded in Stuttgart – Starting with Piano Actions

In October 1882, Louis Renner founded his workshop on Heusteigstrasse in Stuttgart as a dedicated supplier of precision piano actions.

- **Specialized Precision:** Focus on precisely fitted piano actions.
- **Historic Root:** Foundation of the manufacturing heritage united in Thuringia since 1990.

✦ October 1882 • Founding of the Stuttgart Action Workshop

1882

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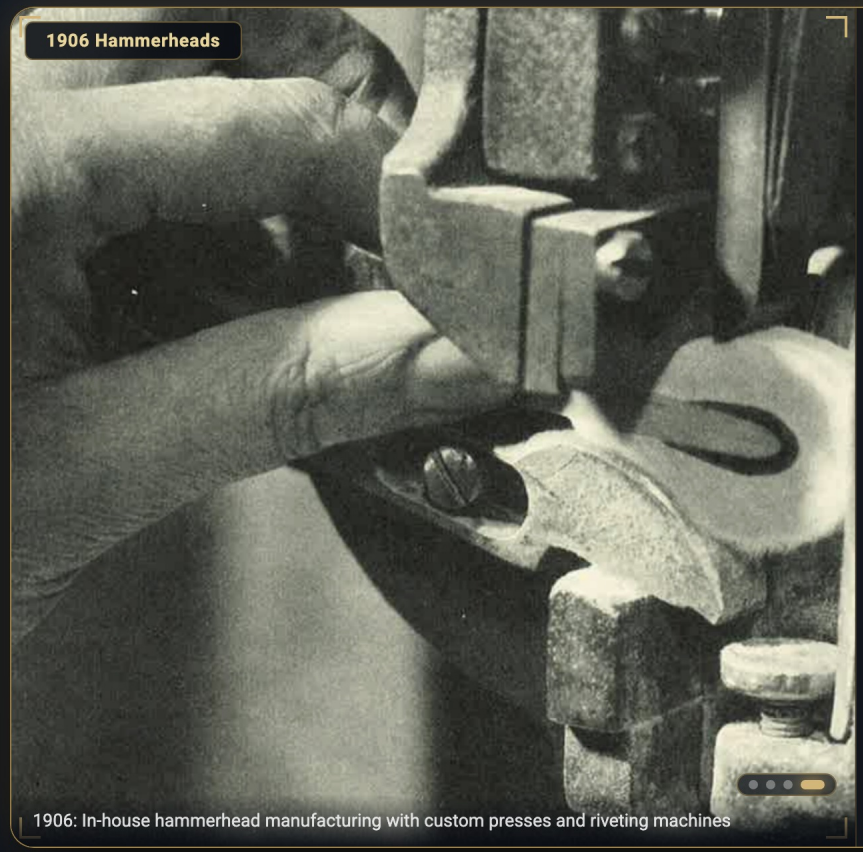
1906 • 2 of 9

1906



1906: In-house hammerhead manufacturing with custom presses and riveting machines

1906

Milestone 2 of 9

1906: Launch of In-House Hammerhead Manufacturing

Starting in 1906, Renner produced its own hammerheads — precisely matched to the geometry and mass of its piano actions.

- **Matched System:** Hammerheads and actions engineered together under one roof for the first time.
- **Custom Presses:** In-house gluing presses ensure consistent felt hoop tension across every set.
- **Concert Benchmark:** Later adopted by Steinway & Sons and leading grand piano makers worldwide.

✦ Since 1906 • In-House Hammerhead Production with Custom Presses

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From a Stuttgart artisan workshop to the global benchmark for piano actions & hammerheads

Electrified workshop with individual machine drives under Oscar Renner (1911)

1911

Milestone 3 of 9

1911: Individual Electric Drives & Grand Piano Actions

In 1911, Oscar Renner launched complete grand piano action manufacturing and converted machinery from belt shafts to hundreds of individual electric motors.

- **Érard Repetition Principle:** Double escapement as the engineering basis of modern grand actions.
- **Complete Portfolio:** Grand actions, upright actions, and hammerheads from a single source.
- **Direct Electric Drives:** Reduced machine vibration and higher repeatability via individual motors.

✦ 1911 • First Renner Grand Actions & Individual Electric Drives

1911

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1944

Historical photo: Destroyed Stuttgart plant in July 1944 prior to reconstruction

1944

Milestone 4 of 9

1944: Wartime Destruction in Stuttgart & Start of Reconstruction

During the air raids of July 24/25, 1944, the Stuttgart factory buildings burned down completely; right afterward, the workforce began rebuilding.

- **July 1944:** Complete loss of the Stuttgart main plant and production machinery.
- **Immediate Restart:** Salvaging usable tooling and rebuilding workshops step by step.
- **Skilled Workforce:** Retained craftsmanship ensured the survival of precision production.

✦ July 24/25, 1944 • Destruction of Main Plant & Rebuilding

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1951/1952 • 5 of 9

Precision manufacturing after returning to the rebuilt Stuttgart headquarters (1955)

1951/1952

Milestone 5 of 9

1951–1955: Dr. Manfred Renner & Reconstruction Completed

In 1951, Dr. Manfred Renner assumed 3rd-generation leadership, completed rebuilding by 1955, and expanded international exports.

- **3rd Generation (1951):** Dr. Manfred Renner takes over management of the family company.
- **Return to HQ (1955):** Full commissioning of the newly rebuilt Stuttgart main factory.
- **International Growth:** Supplying premier upright and grand piano makers worldwide.

✦ 1951–1955 • Dr. Manfred Renner & Rebuilt Stuttgart HQ

1951/1952

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1990 • 6 of 9

1990

Action manufacturing in Zeitz (1990) – Foundation for today's Meuselwitz plant

1990

Milestone 6 of 9

1990: German Reunification & Acquisition of the Zeitz Action Plant

In 1990, Renner acquired Piano Union's action plant in Zeitz – securing skilled specialists and production capacity while laying the foundation for Meuselwitz.

- **Zeitz Acquisition (1990):** Integration of the experienced Piano Union action workforce.
- **Major OEM Capacity:** Reliable supply for key customers including Schimmel (>40 pianos/day).
- **Central German Base:** Direct predecessor to today's Meuselwitz headquarters.

✦ 1990 • Zeitz Acquisition & Long-Term Competitiveness

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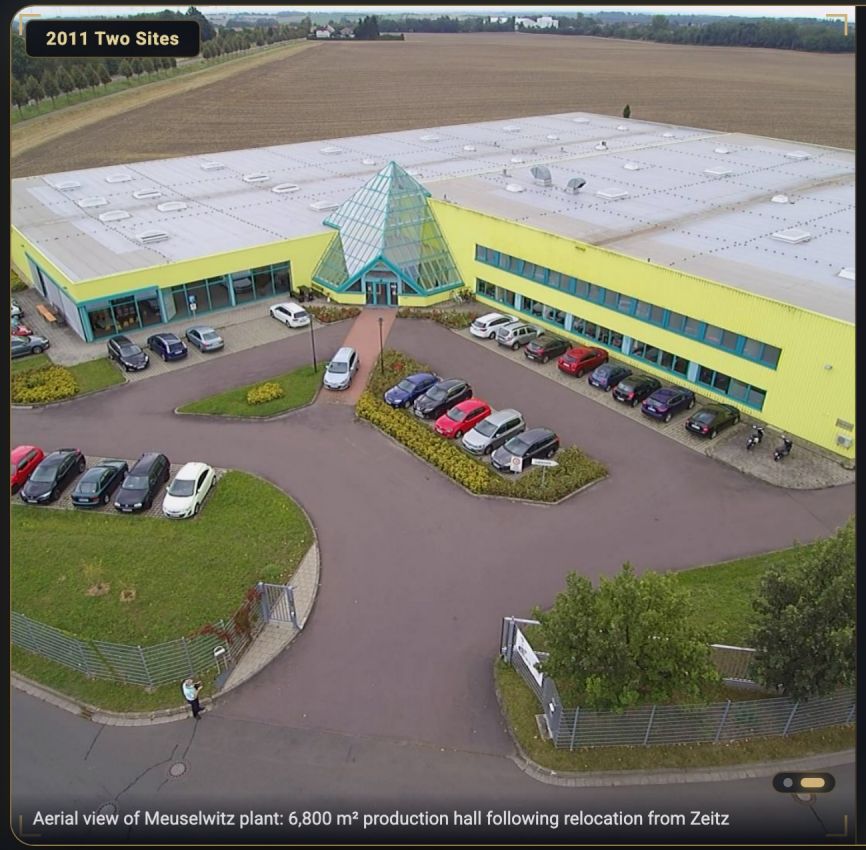
2011 • 7 of 9



Aerial view of Meuselwitz plant: 6,800 m² production hall following relocation from Zeitz

2011

Milestone 7 of 9

2011: Consolidation into Two Core Sites – Gärtringen & Meuselwitz

In 2011, Renner consolidated its previously dispersed workshops into two modern facilities: Gärtringen and Meuselwitz.

- **Move to Meuselwitz:** Relocation from Zeitz into a modern 6,800 m² production hall.
- **Meuselwitz Plant:** Manufacturing of grand/upright actions, hammerheads, and components.
- **Gärtringen Facility:** Specialized woodworking and administration.

✦ 2011 • Two Modern Plants: Gärtringen & Meuselwitz (6,800 m²)

2011

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2019

Clemens von Arnim in production († April 23, 2024): Stewardship for staff & quality

2019

Milestone 8 of 9

2019: Acquisition by Steinway & Sons with Full Operational Autonomy

In 2019, owner Clemens von Arnim entrusted Renner to Steinway & Sons — safeguarding jobs, quality standards, and open supply to piano makers worldwide.

- **Operational Autonomy:** Renner continues to supply all upright and grand piano makers worldwide.
- **Safeguarding Staff:** Long-term industrial stewardship chosen over higher financial bids.
- **In Grateful Memory († 2024):** Honoring Clemens von Arnim for his responsible leadership.

✦ 2019 • Steinway & Sons & In Memory of Clemens von Arnim († 2024)

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Today • 9 of 9

Meuselwitz Headquarters (2026): Drone footage of the modernized plant with new central filtration unit

Today

Milestone 9 of 9

Today: All Manufacturing United at the Meuselwitz Headquarters

Following project launch in October 2025, action, woodworking, and hammerhead production are now fully united under one climate-controlled roof in Meuselwitz.

- **Short Paths Under One Roof:** From timber cutting to finished hammerheads and complete piano actions.
- **Modern Infrastructure:** New central dust filtration plant and year-round hall climate control.
- **Gratitude to Gärtringen:** Deep appreciation to the Gärtringen team for decades of precision work and knowledge transfer.

✦ Meuselwitz Headquarters • Actions & Hammerheads Under One Roof

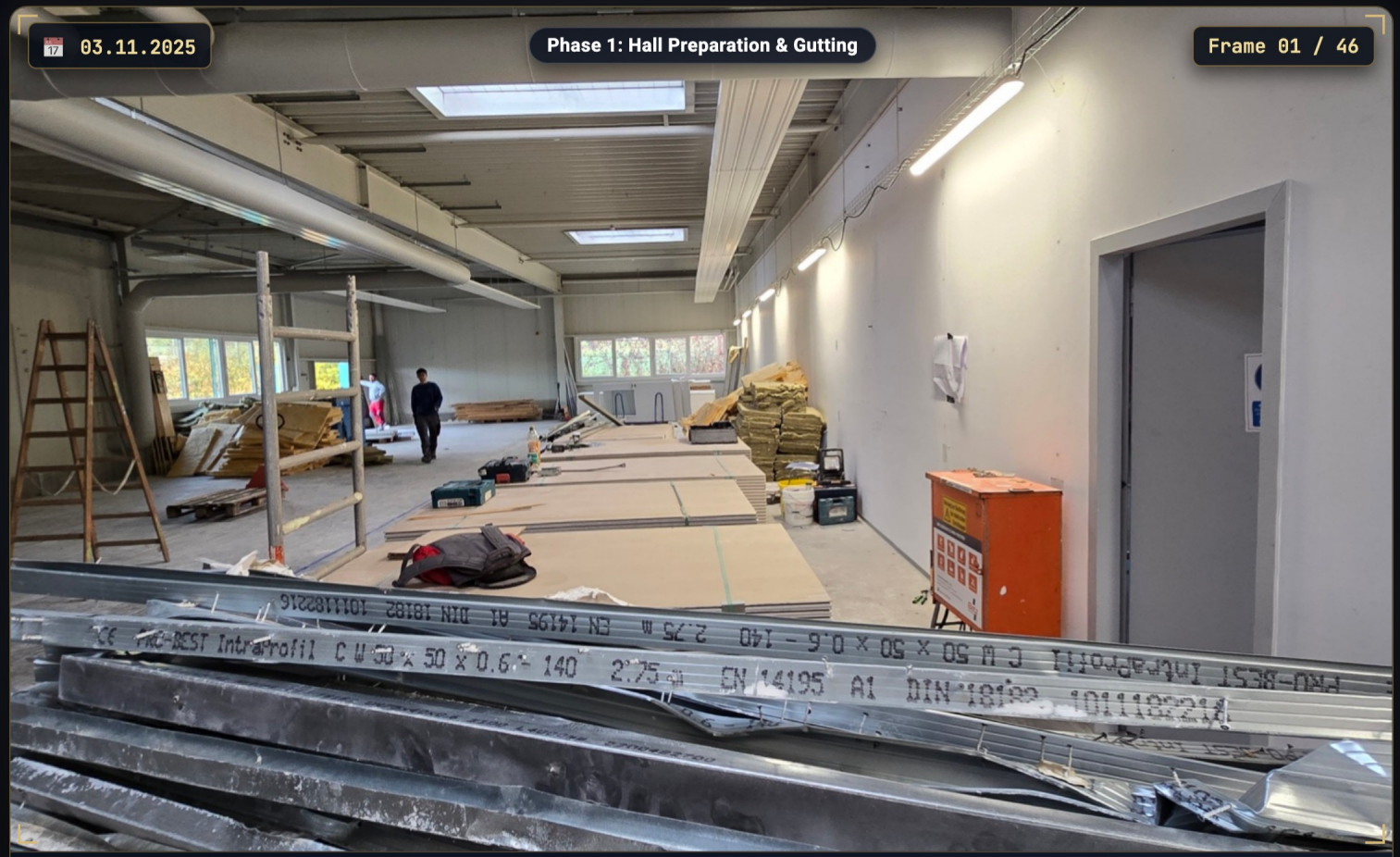
Today

PART 1: INTRODUCTION & HISTORY • MEUSELWITZ PLANT CONVERSION

Construction Documentary: Building the Unified Meuselwitz Plant

Chronological photo journey (11.2025 – 09.2026) followed by an interior drone flight through production

◄ PREVIOUS FRAME



NEXT FRAME ►

Project Kickoff: Site Gutting • Gutting and structural preparation of core hall lines for the future administration and reception wing.

11|25 | 12|25 | 01|26 | 02|26 | 03|26 | 04|26 | 05|26 | 06|26 | 07|26 | 09|26

Drone Flight Finale

PART 1: INTRODUCTION & HISTORY • MEUSELWITZ PLANT CONVERSION

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Foyer & Entrance Area Demolition • View from the foyer toward felt production: cleared large-scale areas for the new manufacturing infrastructure.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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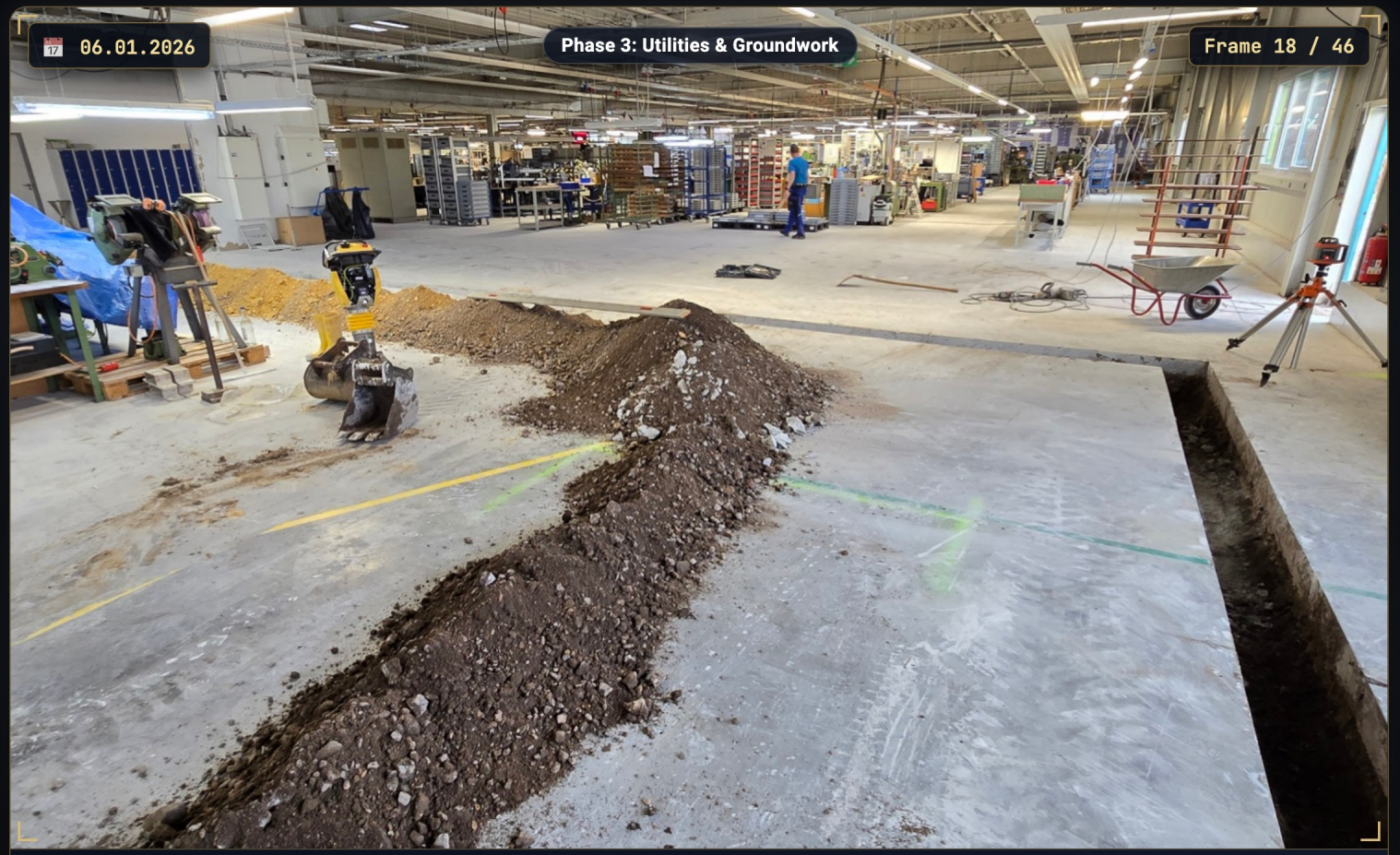

Utilities & Fresh Water Supply • Precision trenching for fresh and industrial water lines – conducted seamlessly without disrupting ongoing production.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Upright Action Assembly Installation • Former warehouse areas are transformed into a state-of-the-art production line for upright piano wippens.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Smart Storage Logistics & Vertical Lifts • Maximum space efficiency: installation of automated vertical shuttle towers storing tens of thousands of precisi...

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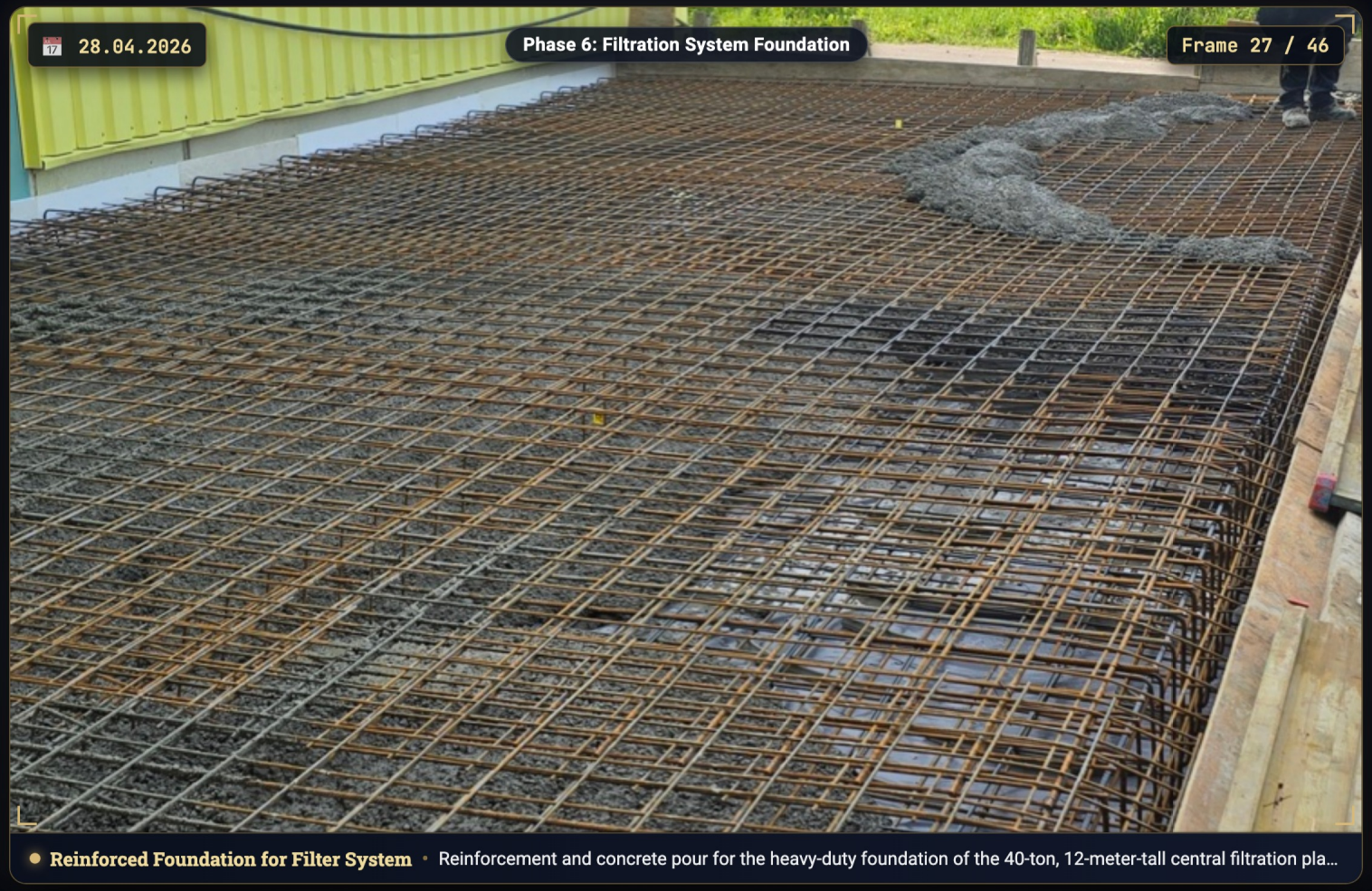

Reinforced Foundation for Filter System • Reinforcement and concrete pour for the heavy-duty foundation of the 40-ton, 12-meter-tall central filtration pla...

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Delivery & Crane Lift of Filter Modules • Heavy-duty logistics with a 70-ton mobile crane: millimeter-accurate positioning of multi-ton filtration components.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Crane Assembly of 12 m Filter Tower • Assembly of the 12-meter-tall upper filter tower section onto the vibration-isolated steel frame.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Relocation of Hammerhead Production • Relocation of hammerhead production into the newly built, climate-controlled hall section in Meuselwitz.

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Assembly of 120,000 m³/h Filter Plant • With a 120,000 m³/h airflow capacity, the new filter plant ensures dust-free, dimensionally stable production.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Extraction Trusses & Hall Ductwork • Installation of large-diameter overhead ductwork for continuous dust extraction across all woodworking machines.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Rooftop Plant: Process Air & HVAC • Year-round constant temperature and 45–50% relative humidity for dimensionally stable wood and felt processing.

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

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Precision Humidity Control for Wood & Felt • Controlled humidification prevents checking in tonewoods and dimensional drift in hammer felt.

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Chronological photo journey (11.2025 – 09.2026) followed by an interior drone flight through production

Molding Line Setup & Full Operation • Final department relocated: all wood, action, and hammerhead manufacturing now operates under one roof in Meu…

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INTERIOR DRONE FLIGHT • MEUSELWITZ PLANT (2026)

The Completed Meuselwitz Plant – Interior Drone Flight

11|25 12|25 01|26 02|26 03|26 04|26 05|26 06|26 07|26 09|26

Drone Flight Finale

PART 2: FELT MANUFACTURING • HISTORY & TRADITION

Filzfabrik Wurzen (FFW) – The Birthplace of Piano Hammer Felt

Wurzen 1783 • From the first piano felt in 1847 to the Weickert Spezial partnership with Renner

1783 WURZEN 1783 | 1847 1ST PIANO FELT | 1850–67 WORLD EXPO | 1861 WURZEN MILL | 1906 WORLD LEADER | Today WEICKERT SPECIAL

ORIGIN WURZEN 1783

1783

Founding of the Felt Manufactory in Wurzen

The world's oldest industrial felt manufactory – rooted in Saxony's Mulde valley

- **Founding in 1783:** Johann Dankegott Weickert establishes the Saxon wool and felt tradition in Wurzen.
- **Fine Felt Pioneer:** The artisan shop grows into a leading center for premium technical virgin wool felts.
- **Piano Focus:** Early dedication to highly resilient felt for the emerging pianoforte industry.

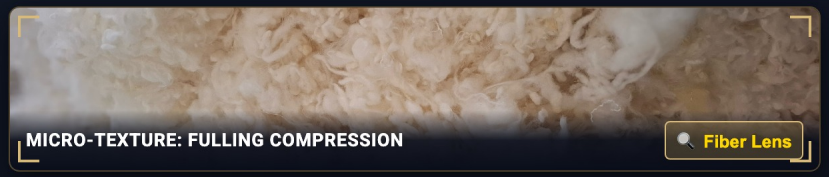
🏛 World's oldest industrial felt manufactory in Wurzen

From hand-carded raw wool fleece to high-density top-felt sheets

100% PURE VIRGIN MERINO WOOL

The Secret of Weickert Felt

Purely mechanical fulling without chemical additives – for rapid rebound and clear concert tone



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- **Pure Fulling without Resins:** Densification solely through Mulde river water, heat, and pressure – zero dampening resins or glues.
- **Permanent Elastic Springback:** Natural keratin springs rebound in milliseconds – liberating the entire overtone harmonic spectrum.
- **Proximity to Meuselwitz (65 km):** Only ~65 km distance ensures direct exchange between felt and hammer masters and immediate availability.

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INVENTION 1847

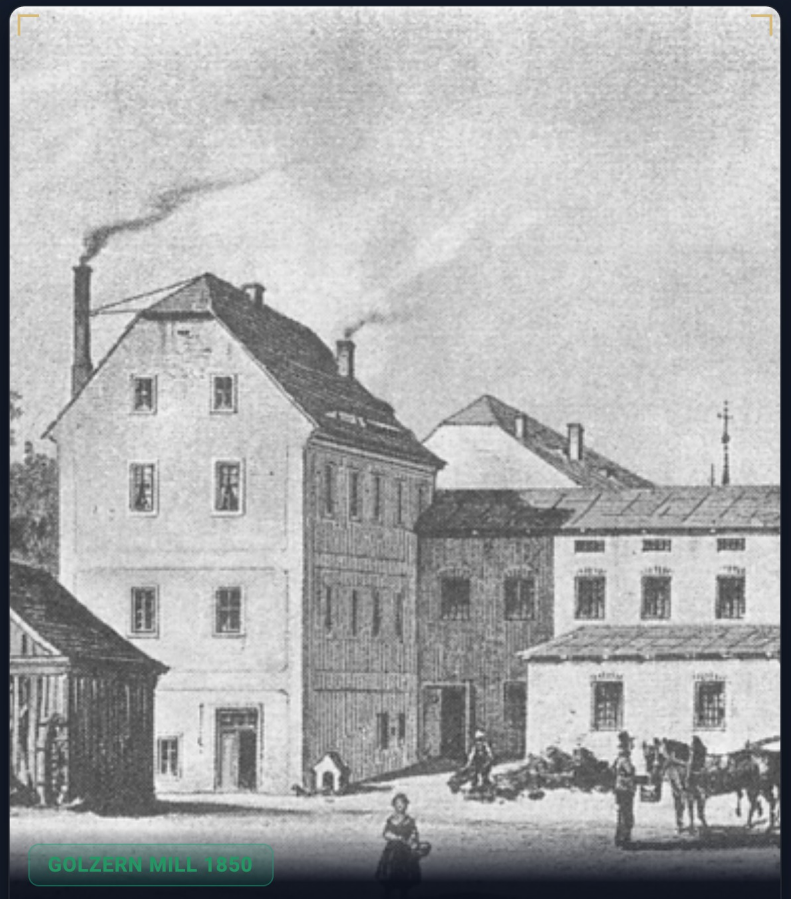
1847

Germany's First Viable Piano Hammer Felt

Patzschke & Weickert revolutionize European piano craftsmanship in 1847

- **Felt Replaces Leather:** In 1847, uneven leather hammer coverings are successfully replaced by homogeneous felt.
- **Strong Partnership:** Weickert's enterprise and Patzschke's felt expertise establish international renown.
- **Soft River Water:** The naturally soft, low-mineral Mulde water ensures pristine, resilient wool fibers.

🎹 **Milestone: Wurzen felt permanently replaces leather in pianos**

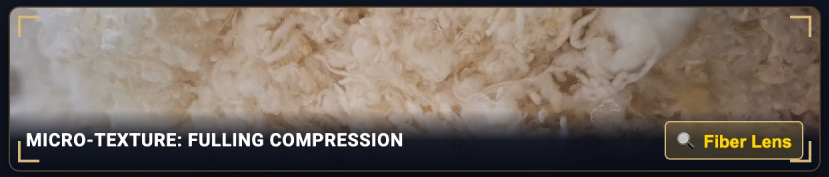


Golzernmühle near Grimma (1850): Early felt making on an island in the Mulde river

100% PURE VIRGIN MERINO WOOL

The Secret of Weickert Felt

Purely mechanical fulling without chemical additives – for rapid rebound and clear concert tone



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LONDON & PARIS 1851

1851

World Expos & Gold Medals in London & Paris

International gold medals at the World Exhibitions in Leipzig, London, and Paris

- **World Expo Gold:** Top honors in Leipzig (1850), London (1851), and Paris (1867) establish global acclaim.
- **Royal Visits:** Saxon monarchs personally inspect and praise the superlative craftsmanship of the Wurzen mills.
- **Europe's Top Choice:** Premier piano builders across Vienna, Paris, and London adopt Weickert felt exclusively.

🏅 Gold Medals London 1851, Paris 1867 & Leipzig 1850

AWARDS & MEDALS

Historical World Expo medals awarded for superlative felt excellence

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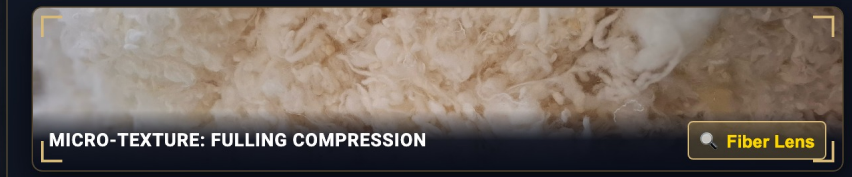

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EXPANSION 1861

1861

Industrial Scale at Mühlgraben in Wurzen

Steam power, modern fulling mills, and the clean waters of the Mulde river

- **Industrial Scale:** In 1861, modern facilities with proprietary steam power are erected at Wurzen's Mühlgraben.
- **Precision Fulling Mills:** Steam-powered machinery guarantees unprecedented density and batch uniformity.
- **Gentle Wool Scouring:** Mineral-poor river water preserves the natural cuticle scale structure of the Merino wool.

⚙ Steam fulling & natural river water at Wurzen Mühlgraben

Original historical engraving of Pianofortefilzfabrik in Wurzen (1871)

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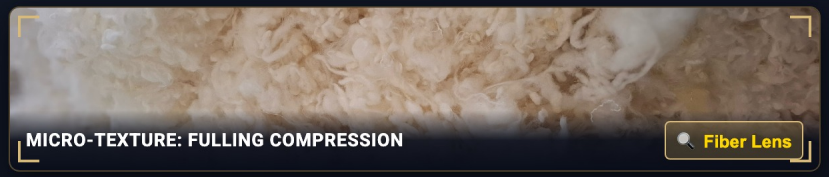

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WORLD LEADER 1906

1906

Global Market Leader for 200,000 Pianos Annually

Director Patzschke, his master craftsmen & 350 specialists at Wurzen (1906)

- **Market Leader by 1906:** Wurzen equips up to 200,000 pianos and grands worldwide with hammer felt each year.
- **Global Benchmark:** Steinway & Sons New York and Europe's premier manufacturers rely on Wurzen felt.
- **Masters & Director Patzschke:** Historic 1906 photograph standing before the monumental tapered felt block.

🌎 Global reference standard for Steinway & Sons New York

Original historical photograph (Sept. 1906): Master felters of Filzfabrik Wurzen with Director Patzschke at the felt cutting table

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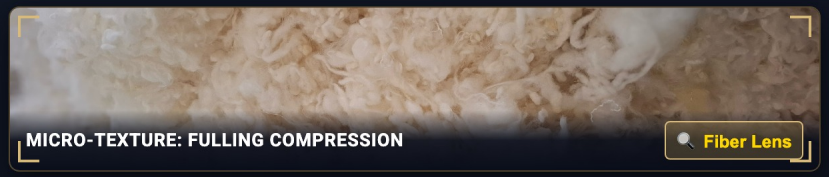

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- **Permanent Elastic Springback:** Natural keratin springs rebound in milliseconds – liberating the entire overtone harmonic spectrum.
- **Proximity to Meuselwitz (65 km):** Only ~65 km distance ensures direct exchange between felt and hammer masters and immediate availability.

PART 2: FELT MANUFACTURING • HISTORY & TRADITION

Filzfabrik Wurzen (FFW) – The Birthplace of Piano Hammer Felt

Wurzen 1783 • From the first piano felt in 1847 to the Weickert Spezial partnership with Renner

1783 WURZEN 1783 | 1847 1ST PIANO FELT | 1850–67 WORLD EXPO | 1861 WURZEN MILL | 1906 WORLD LEADER | Today WEICKERT SPECIAL

TODAY • RENNER PARTNER

Today

Renaissance of "Weickert Spezial" & Partnership with Renner

Historic Wurzen mill & direct geographical proximity to Renner's factory in Meuselwitz

- **Revived in 2010:** The Brand family restores the historic 1903 Wurzen formula for concert-grade hammer felt.
- **Geographical Proximity (~65 km):** Wurzen and Renner's Meuselwitz plant share Central Germany's piano craft cluster.
- **Direct Master Collaboration:** Short travel times enable face-to-face calibration of felt density and elasticity on-site.

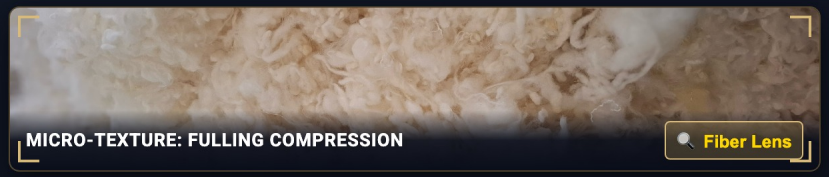
📍 **Only 65 km: Direct partnership Wurzen ↔ Renner Meuselwitz**

Wurzen Weickert Spezial felt: Flawless voicing response & tonal depth

100% PURE VIRGIN MERINO WOOL

The Secret of Weickert Felt

Purely mechanical fulling without chemical additives – for rapid rebound and clear concert tone



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PART 2: FELT MANUFACTURING • RAW WOOL & FORMULA

Pure Merino Wool & the Secret of Fiber Blending

Uncompressed overseas raw wool and tailored fiber recipes as the foundation of hammerhead felt

Stage 01

RAW WOOL SELECTION

Uncompressed Overseas Merino Wool & Intact Fiber Structure

- **Uncompressed Bale Delivery:** Loosely packed shipping preserves the microscopic keratin scale structure intact.
- **Pronounced Fiber Crimp:** Millions of natural micro-springs ensure optimal interlocking during fulling and rapid rebound.
- **Natural Lanolin:** Pure wool grease and gentle scouring keep the keratin fibers supple for decades.

Loosely delivered Merino wool retaining its strong natural fiber crimp

Stage 02

FIBER FORMULA

Gentle Fiber Opening & Calibrated Blend Formula

- **Calibrated Fiber Blend:** Precise balance of long tensile fibers (for the crown) and fine filler wool (for core density).
- **Multi-Stage Opening:** Gentle plucking and pneumatic aeration separate fiber tufts cleanly into individual fibers.
- **Climate Conditioning:** Controlled moisture uptake prevents static charge, fiber breakage, and density spots.

Pneumatic loosening and blending of wool batches at FFW Wurzen

PART 2: FELT MANUFACTURING • CARDING & FIBER ALIGNMENT

Carding: Parallel Fiber Alignment into Fine Fleece Webs

Defined fiber orientation as the basis for uniform elasticity and predictable needle voicing

FIBER ORIENTATION

03

Drum cards comb fibers in parallel alignment

Deliberate longitudinal and transversal alignment for predictable strike elasticity

- **Carding:** Fine wire rollers gently comb the opened wool, untangling and aligning the fibers in parallel.
- **Defined Fiber Course:** The gossamer-thin fleece web receives a precisely controlled primary fiber direction.
- **Controlled Layering:** Superimposed fleece webs build target sheet thickness, tensile strength, and wedge taper.

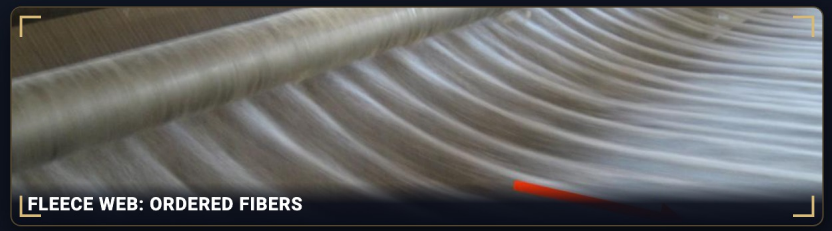
🧵 **Parallel fiber guidance is the cornerstone of precise voicing**

Carding Merino wool into uniform fleece webs with defined fiber alignment

FLEECE STRUCTURE

Prerequisite for Voicing

Only parallel aligned fibers respond predictably and uniformly to needle penetration during voicing



- **Gliding Needling:** Voicing needles part parallel fibers rather than severing them.
- **Fluid Density Gradients:** Cross-layered stacking prevents delamination under extreme touch forces.
- **Maximum Fiber Length:** Gentle carding prevents fiber breakage and ensures maximum tensile strength of the fleece.

PART 2: FELT MANUFACTURING • THE FULLING PROCESS

The 4 Pillars of Fulling: Steam, Pressure, Motion & Experience

Purely mechanical-thermal fiber interlocking with soft Mulde water – 100% free from glues and resins

HAMMER MILL

04

Pure mechanical felting without extraneous additives

Water, heat, pressure, and rhythm permanently interlock fibers

- **1. Live Steam:** Opens microscopic cuticle scales of fibers and renders keratin supple.
- **2. Mechanical Pressure:** Continuously compresses fleece layers, initiating progressive density increase.
- **3. Oscillating Motion:** Frictional rubbing interlocks natural barbed scales of fibers inextricably.
- **4. Artisan Know-How:** Sensory feel for the exact moment of completion governs elasticity and acoustic resonance.

🌊 **Solely soft Muldental water and heat – 100% chemical-free**

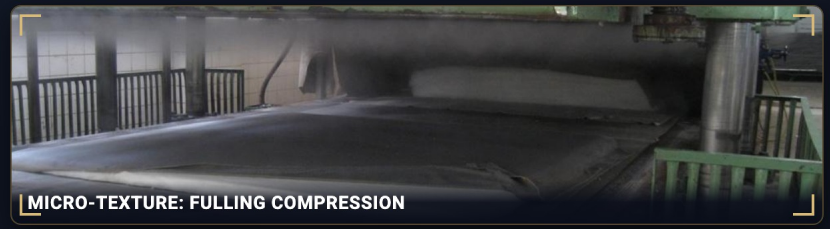
CRAFTSMANSHIP

Fulling layered fleece webs under steam, pressure, and motion into a homogeneous felt sheet

FULLING COMPRESSION

Physical Compaction

Scales of wool hairs interlock micromechanically like burrs



MICRO-TEXTURE: FULLING COMPRESSION

- **No Damping by Glues:** Pure mechanical felting preserves free oscillation of wool fibers.
- **Permanent Elasticity:** Natural keratin springs rebound within milliseconds without fatigue.
- **Maximum Overtone Purity:** Soft Muldental water prevents mineral deposits inside fiber pores.

PART 2: FELT MANUFACTURING • MAXIMUM FULLING DENSITY

56% Fulling Density (0.56 g/cm³): The Natural Limit of Mechanical Fulling

Maximum natural springback without fiber damage or chemical over-hardening

PHYSICAL LIMIT

56 %

Full Densification at the Physical Limit

Felt sheets are compacted precisely to the point where fibers contact optimally

- **Maximum Fiber Compression:** Fibers cannot be mechanically compacted further without structural damage.
- **Peak Natural Spring Rate:** Corresponds to the physical optimum of inherent elasticity for concert grand touch.
- **Free from Synthetic Resins:** Eliminating chemical hardeners ensures permanent organic tone blooming.

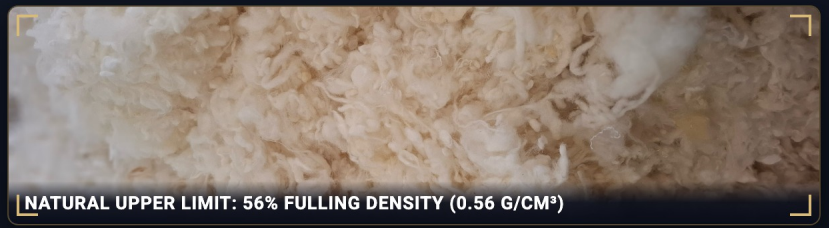
🎯 **Max. 56% Fulling Density (0.56 g/cm³): Peak elasticity with zero hardeners**

Post-fulling finishing of the mechanically fulled felt sheet in Wurzen

ACOUSTIC PERFECTION

Elastic Springback Instead of Brittle Hardness

56% natural fulling density delivers brief string contact times and nuanced dynamic control.



NATURAL UPPER LIMIT: 56% FULLING DENSITY (0.56 G/CM³)

- **Rapid String Rebound:** The hammer rebounds instantly off strings without muting tone.
- **Color-Rich Tone Spectrum:** Full blooming of fundamentals and overtones without treble harshness.
- **Stable Hammer Action:** Consistent density sustained over tens of thousands of keystrokes.

PART 2: FELT MANUFACTURING • FINAL COMPACTION & LAYER ANATOMY

Final Compaction & Tapered 36-Layer Anatomy

Dimensional calibration without sanding – 2×6 continuous outer webs preserve full bending tension

SHAPING WITHOUT GRINDING

05

Manual compaction and leveling

Targeted hammer blows bring felt sheets to exact size while relaxing the fiber lattice.

- **Mechanical Hammering:** Felt hammers and polished natural stones consolidate sheets to exact dimensions.
- **Zero Sanding Removal:** The Weickert process eliminates sanding completely—not a single tension fiber is cut.
- **Unbroken Bending Radius:** When pressed around the hammer moulding, the outer skin stays smooth and tear-free.

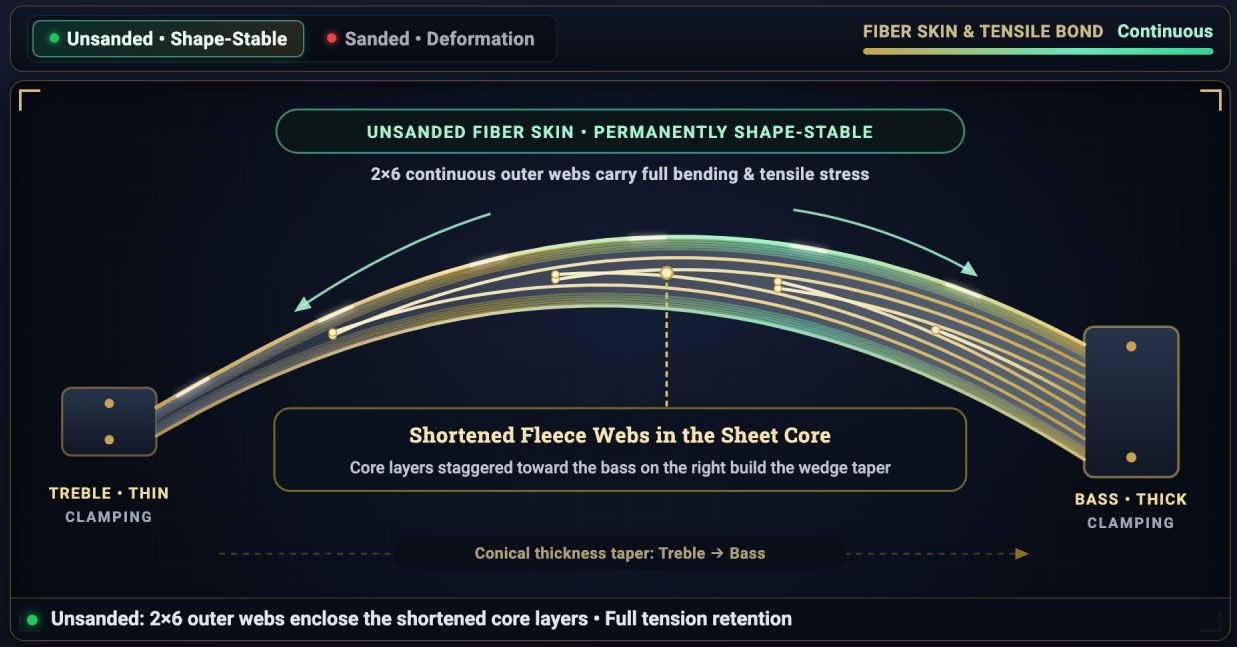
Intact surface fiber skin ensures maximum longevity

Fulled felt sheets with natural indentation firmness of 56 to 68 Shore-A

CONICAL FLEECE STRUCTURE & BACKBONE

Conical Taper & Tensile Stability: The Fleece Web Backbone

From thin treble on the left to thick bass on the right, shortened core webs form the wedge—enclosed by 2×6 outer webs.



- **Treble → Bass:** Shortened core webs build the thick bass on the right while keeping the treble thin on the left.
- **2×6 Outer Webs:** Continuously unsanded—absorbs all bending and tensile forces.
- **Sanding Effect:** Severed outer webs slip apart—loss of structural pre-tension.

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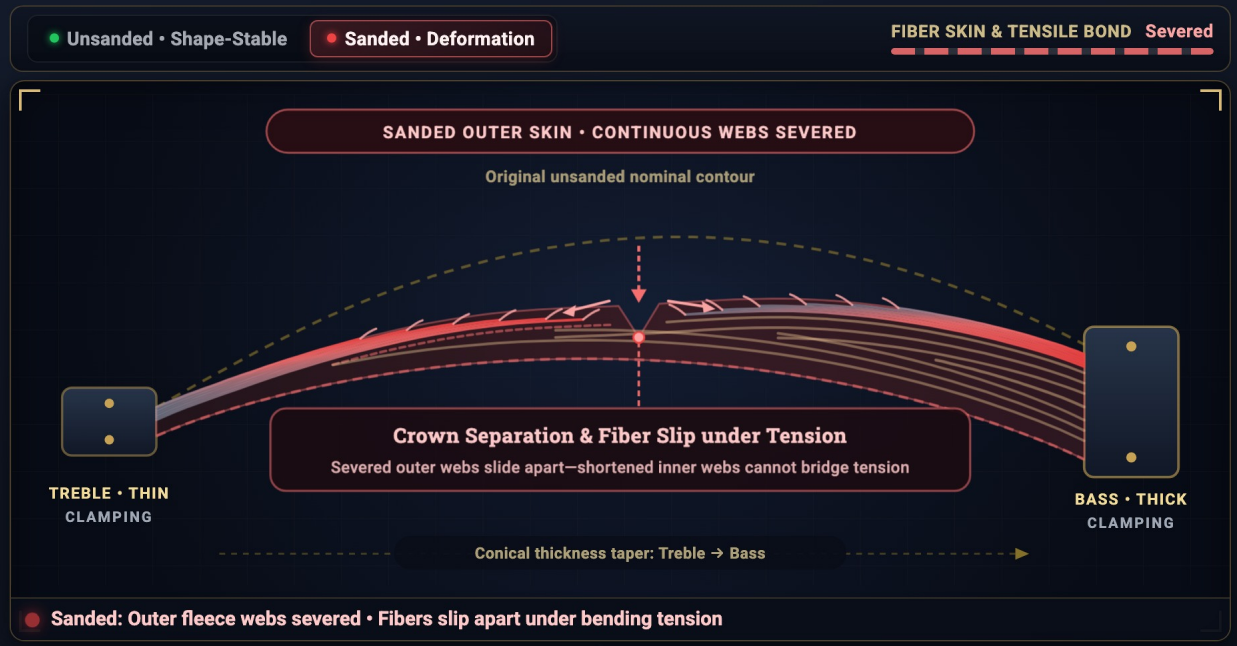

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PART 3: HAMMERHEAD PRODUCTION · FELT STORAGE & ASSORTMENT

Climate-Controlled Felt Storage & Assortment Diversity

Over 30 tapered top-felt formulas plus 6 standard underfelt colors and custom dyeings in stock

30+ Grades

TOP-FELT INVENTORY

Tapered FFW & Weickert Sheets in Controlled Seasoning

- **Mill-Fulled Taper:** Thick in the bass, progressively slimmer and denser toward the treble.
- **Broad Assortment:** Over 30 calibrated FFW and Weickert formulas for concert, OEM, and historic instruments.
- **Climate & Resting Period:** Regulated humidity, pheromone monitoring, and seasoning relieve fulling stresses.

TOP-FELT STORAGE

Drone flight through the climate-controlled felt sheet storage in Meuselwitz

CLASSIC

Red & Wine Red

Traditional Renner signature colors for European concert and prestige grands.

HERITAGE & CONCERT

Orange

Classic specification for concert instruments and period-authentic restorations.

MANUFACTORIES

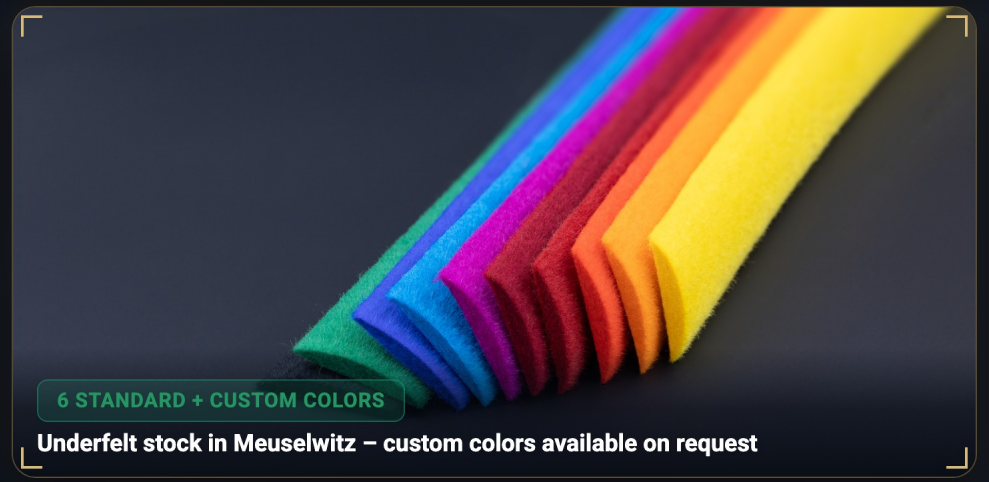
Yellow

Heritage underfelt color of renowned Saxon and Viennese piano makers.

MARQUES & CUSTOM

Blue, Black & Custom

Classic blue (e.g., C. Bechstein), sleek black, plus bespoke customer dyeings.



6 STANDARD + CUSTOM COLORS

Underfelt stock in Meuselwitz – custom colors available on request

PART 3: HAMMERHEAD PRODUCTION · WEICKERT SPECIAL & SEM MICROSTRUCTURE

"Weickert Special" – Exclusive to Renner

56% maximum fulling density and elevated residual wool grease per the original 1903 Wurzen formula

56% Fulling Density

FFW WURZEN × STFI LAB REPORT

Highest Fulling Density & Protected Keratin

Gentle wool scouring and intensive fulling unite 56% peak density with elevated wool grease retention (STFI Report P2024 0903).

56%	0.36%	−65%
Max. Fulling Density	**Residual Grease (ISO 3074)**	**Climate-Cycle Swelling**
0.56 g/cm³ chemical-free	+38% vs. 0.26% sample	4.81% vs. 13.69%

- **56% Fulling Density:** Highest purely mechanical density (0.56 g/cm³) for strong internal tension without chemical hardeners.
- **0.36% Residual Grease (ISO 3074):** Gentle scouring retains +38% more natural wool wax (vs. 0.26%) as moisture protection.
- **Verified Climate Stability:** In the 48h STFI climate-cycling test, thickness swelling drops by 65% (only 4.81% vs. 13.69%).
- **Intact Cuticle & Voicing Stability:** Epicuticle scales visible under SEM ensure strong fiber traction and lasting tonal consistency.

Verified: Accredited STFI Test Report P2024 0903 (M. Kermer, ISO 3074 & Climate Cycle)

1× Overview (1 mm) | 10× Fiber Detail (100 µm) | 22× Cuticle Zoom | Live Loupe | Felt Strip W 1512 K

STFI • SEM

SEM Overview (STFI Mik59_24 • 1 mm): Ultra-dense 3D fiber interlock with 56% maximum fulling density.

Move cursor for 10× live loupe • Click or scroll to zoom

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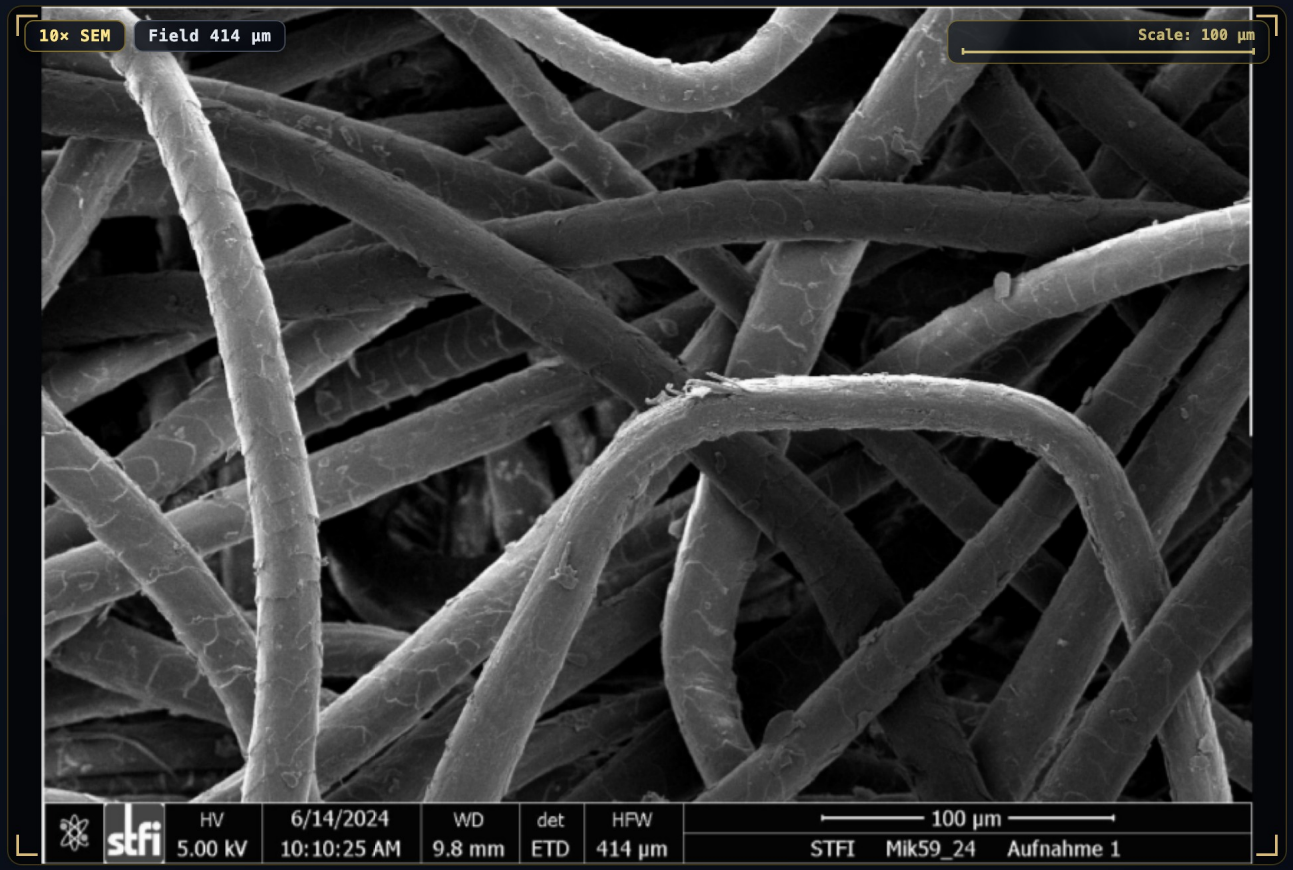

STFI • SEM

SEM Detail 10× (STFI Mik59_24 • 100 µm): Merino fibers with intact cuticle scales and +38% higher residual wool grease (0.36%).

Click for 22× cuticle zoom • Scroll for continuous zoom

PART 3: HAMMERHEAD PRODUCTION · WEICKERT SPECIAL & SEM MICROSTRUCTURE

“Weickert Special” – Exclusive to Renner

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STFI • SEM

SEM Deep Zoom (>18×): Overlapping epicuticle scales and lanolin-protected keratin ensure fiber traction & climate stability.

Drag to pan • Scroll wheel or slider for 1×–25× zoom

PART 3: HAMMERHEAD PRODUCTION · FELT PREPARATION

Felt Preparation: Top-Felt Cutting & Underfelt Profiling

Robotic strip cutting to 1 mm tolerance and precision profile milling prior to pressing

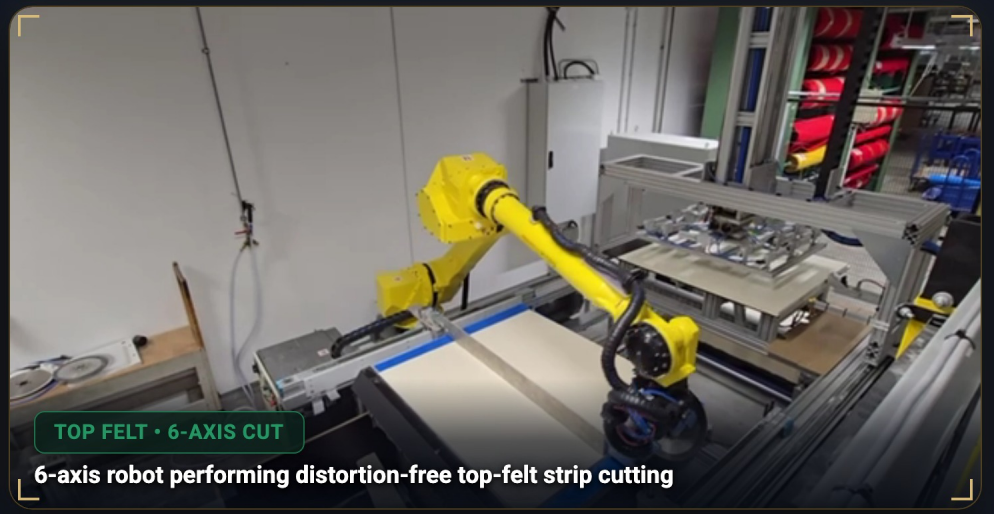
≤ 1 mm Tolerance

01 · 6-AXIS ROBOTICS

Robotic Top-Felt Cutting

- **1 mm Strip Cutting:** 6-axis robot divides seasoned tapered felt sheets distortion-free across the full width.
- **Crush-Free Cold Cut:** Sharp blades part the Merino fibers cleanly without thermal glazing or edge compression.
- **Inline Batch Coding:** Sheet and order IDs marked directly on every strip for complete traceability.

03



TOP FELT · 6-AXIS CUT

6-axis robot performing distortion-free top-felt strip cutting

± 0.1 mm Tolerance

02 · PROFILE MILLING

Underfelt Profile Milling

- **Elastic Coupling Bridge:** Contour-milled underfelt transfers strike energy smoothly from the rigid hammer moulding into the top felt.
- **Feathered Tails (± 0.1 mm):** Ends milled to a ± 0.1 mm tolerance ensure a seamlessly closed hammer contour right into the high treble.
- **Balanced Press Load:** The calibrated wedge contour distributes jaw pressure evenly across both hammer flanks.

04

UNDERFELT · PROFILE MILLING

Contour-milling the underfelt strip before loading into the glue press

PART 3: HAMMERHEAD PRODUCTION · THE HAMMER MOULDING

The Hammer Moulding: Tonewoods, Properties & Profiles

From heavy hardwoods (hornbeam, maple) and oil-rich Kotibé to lightweight, fine-grained American walnut

HARDWOOD · IDEAL — Heavy · Stable

Hornbeam (Weißbuche)

Very demanding to season—yet once properly dried, an ideal and uniform moulding wood.

HARDWOOD · ELASTIC — Heavy · Selected

Maple (Ahorn)

Heavy, often more elastic hardwood—though irregular sugar deposits carry a higher risk of variance.

PREMIUM · TREBLE — Very Light · Fine-Grained

American Walnut

Highest-grade wood: tight growth rings, long-fibered, and very light—ideal for a brilliant treble.

SPECIALTY · OIL-RICH — Heavy · Specialty

Kotibé (Mahogany)

Comparatively heavy due to its high natural oil content—suited only for specific applications.

FORM B · US · CSB

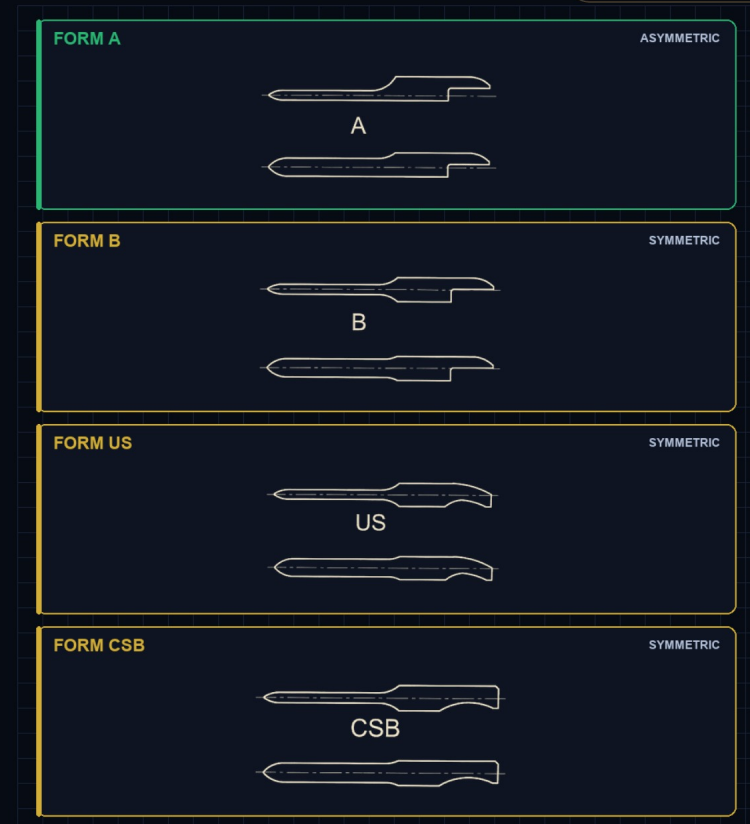
Symmetrical Profiles

Steinway & Sons, Bösendorfer, C. Bechstein, Yamaha, Kawai, Schimmel, Blüthner, Förster, and Sauter.

FORM A

Asymmetrical Profiles

Form A with offset moulding shoulder for vintage Bechstein grands and slender treble scale designs.



PROFILE DRAWINGS

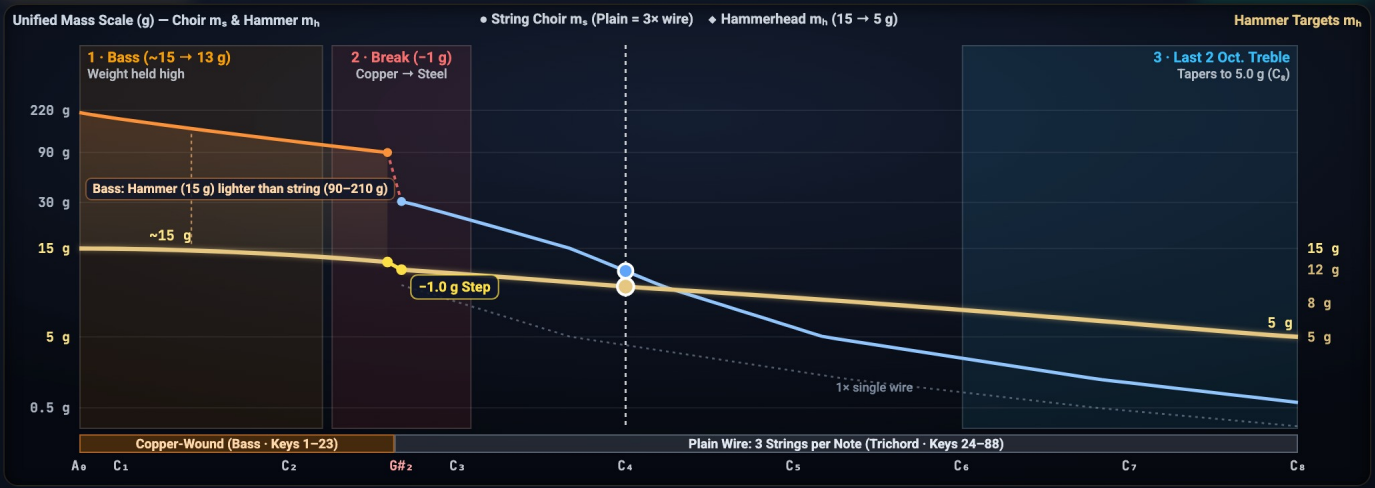
Renner hammer moulding cross-sections: Form A, Form B, Form US, and Form CSB

PART 3: HAMMERHEAD PRODUCTION · ACOUSTIC MASS SCALING

Optimal Hammerhead Weight Relative to String Weight

From ~15 g in the bass across the −1 g break step to 5 g in the treble: balancing contact time and string deflection

All Registers | ① Bass (~15 g) | ② Break (−1 g) | ③ Treble (to 5 g) | Test Strike

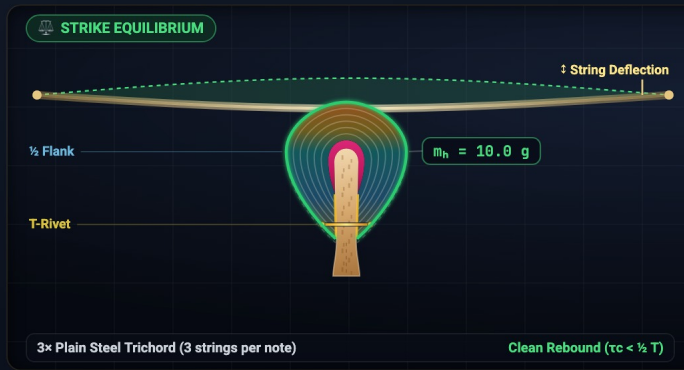


Key 40 · C_4 (262 Hz · Plain Wire · 3 strings/not… String m_s ≈ 11.8 (3×3.9) g | Hammerhead m_h = 10.0 g (Ideal)

STRIKE PHYSICS

Crown Equilibrium: Contact Time τc ↔ String Deflection

Select key A_0–C_8



Crown Contact Time τc — Optimal balance

String Deflection (Energy) — Full singing deflection

Tonal Fullness & Overtone Freedom — Full spectrum & projection

CORE EQUILIBRIUM

String Weight ↔ Hammer Mass

When **string weight** and **hammerhead weight** match, **contact time τc** and **string deflection** reach ideal equilibrium.

Simulate Hammerhead Weight — 0% · Optimal Equilibrium

Too Light | Equilibrium | Too Heavy

Contact Time τc	String Deflection	Mass Balance
τc ≈ 1.6 ms	100% Deflection	Equilibrium
½ String period: 1.9 ms	Full singing deflection	τc ↔ deflection ideal

✦ MASS MATCHING ACROSS ALL 88 KEYS

From ~15 g in the Bass Across the 1 g Break Step to 5 g in Treble

Hammerhead weight must balance actual string weight: bass hammers start around 15 g and are held high as long as possible (15 → 13 g), step down by 1 g onto plain steel in the middle, and finish at 5 g in the treble.

Renner Engineering: Targeted moulding tonewoods (hornbeam to lightweight American walnut), conical Weickert felt taper, precision moulding profiling, and contoured tail shaping.

PART 3: HAMMERHEAD PRODUCTION · GLUE PRESS

Pressing & Gluing to 0.1 mm

Building outer tensile tautness and inner cushion support in the glue press

0.1 mm Tolerance

GLUE PRESSING

Pressing in the Kettle and Heated Side Jaws

Precision glue presses unite top felt, underfelt, and hammer moulding with repeatable accuracy.

- **0.1 mm Pressing Tolerance:** Precision-guided jaws maintain exact dimensional accuracy across the entire set.
- **Kettle & Side Jaws:** The felt is pressed downward into the kettle; the side jaws then close in and press the felt firmly onto the glue surface.
- **Defined Stress Zones:** High outer tensile force meets a compressed, resilient inner cushion.

⚡ This internal tension gradient underpins dynamic range from pp to ff

05

FACTORY VIDEO

Pressing into the kettle and closing the side jaws in Meuselwitz

PART 3: HAMMERHEAD PRODUCTION · CNC MILLING & CALIBRATION

CNC Milling & Robotic Calibration

0.1 mm moulding milling, post-gluing trim milling, and robotic calibration of every individual hammerhead

0.1 mm Tolerance

CNC MILLING & ROBOTICS

Two-Stage CNC Milling & Single-Head Calibration

CNC mills first shape the hammer mouldings and trim the felt overhang after gluing—before a robotic cell calibrates each individual head.

- **1. Moulding Milling (0.1 mm):** CNC milling machines first machine the hammer mouldings to a 0.1 mm tolerance.
- **2. Post-Gluing Trim Milling:** After pressing and gluing, the felt overhang is removed and cleanly finish-milled.
- **3. Robotic Calibration:** Subsequently, a robotic cell precisely calibrates each individual hammerhead.

06

0.1 mm CNC accuracy combined with repeatable robotic single-head calibration

FACTORY VIDEOS

CNC milling center & subsequent single-head calibration on the robotic system

PART 3: HAMMERHEAD PRODUCTION · CNC MILLING & CALIBRATION

CNC Milling & Robotic Calibration

0.1 mm moulding milling, post-gluing trim milling, and robotic calibration of every individual hammerhead

0.1 mm Tolerance

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06

0.1 mm CNC accuracy combined with repeatable robotic single-head calibration

FACTORY VIDEOS

CNC milling center & subsequent single-head calibration on the robotic system

PART 3: HAMMERHEAD PRODUCTION · RIVETING & BORING

Automated Riveting & Boring

Action-matched mounting geometry and permanent felt-tail retention

ø 5.4 mm · Riveting

FINAL FINISHING

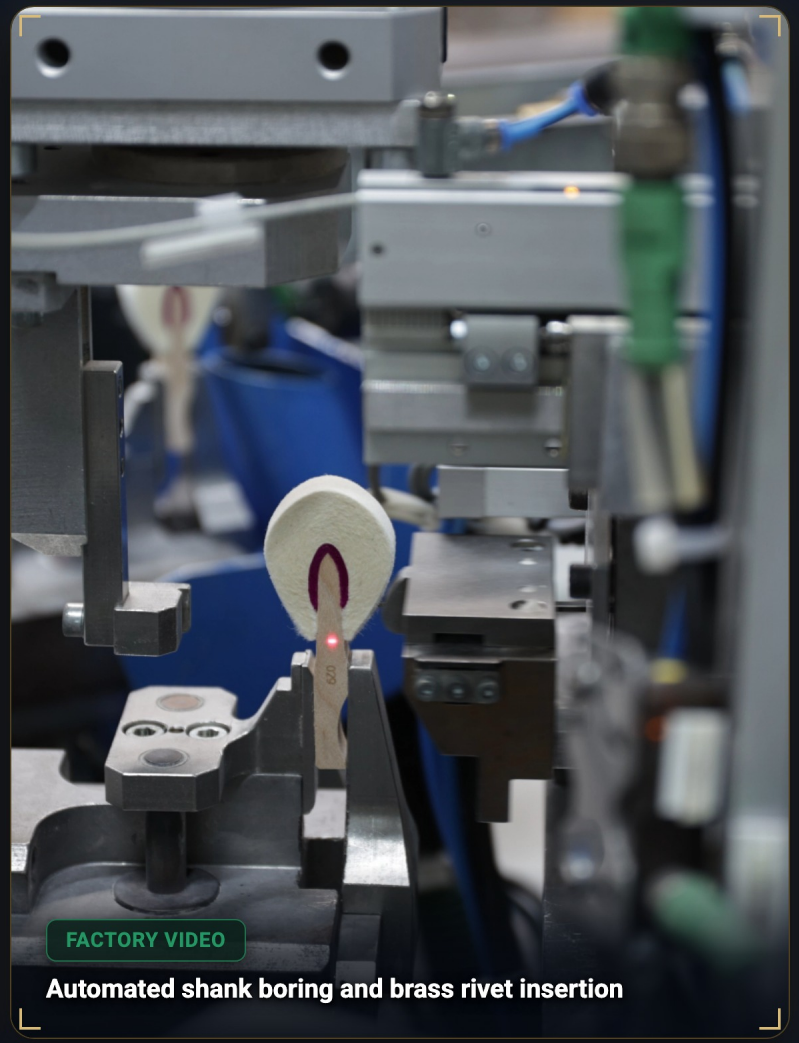
Mounting Geometry for the Hammer Shank

Every individual head receives its model-specific bore geometry and mechanical lock.

- **Precision Boring Angle:** Exact depth, diameter (e.g., ø 5.4 mm), and individual boring angle per note.
- **Brass Riveting:** Permanently locks the bonded felt tails against opening under heavy concert playing.
- **Radiused Hammer Tail:** Cleanly contoured tail radii ensure reliable checking against the backcheck felt.

07

Ready for accurate gluing onto the hammer shanks



Automated shank boring and brass rivet insertion

PART 4: YAMAHA KIT · PRODUCT SHOWCASE

Louis Renner Yamaha Kit (C1–C3) – Seamless Drop-In

Tailored for Yamaha C1–C3: Dynamic expressiveness, brilliant overtones & deep fundamental bass.

MEASUREMENT-BASED PRODUCTION · 5 YEARS R&D

SYSTEM

Proprietary Measurement & Production System

Supreme voicing culture for new and existing Yamaha C1–C3 grand pianos.

- **Calibrated System:** 5 years of R&D for perfectly tuned felt density, elasticity, and tension.
- **Defined Target Tone:** Every set manufactured to millimeter precision for the C1–C3 acoustic optimum.
- **Tone Without Voicing:** Instant full, warm concert sound – zero manual pre-voicing needed.

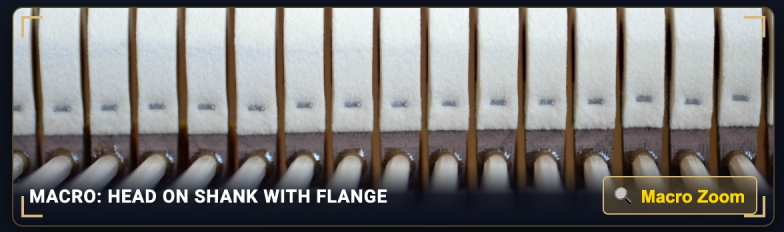
✦ Reliable Concert Tone: Instant master result straight from the factory

Calibrated hammer set: Complete harmonic spectrum without voicing

RENNER YAMAHA KIT

Heads & Shanks in Detail

100% drop-in pre-assembled for Yamaha C1, C2, and C3.



- **Calibrated System:** Proprietary measurement process for finished sound straight from the factory.
- **Walnut & Hornbeam:** Minimized moment of inertia for lightning-fast repetition and agile response.
- **Drop-In & Hide Glue:** Rock-hard acoustic coupling, factory pre-glued – 0 h drilling labor.
- **1-Screw Installation:** Plug & play into original rail – installation & regulation in 4–5 hours on site.

PART 4: YAMAHA KIT • PRODUCT SHOWCASE

Louis Renner Yamaha Kit (C1–C3) – Seamless Drop-In

Tailored for Yamaha C1–C3: Dynamic expressiveness, brilliant overtones & deep fundamental bass.

OPTIMIZED INERTIA

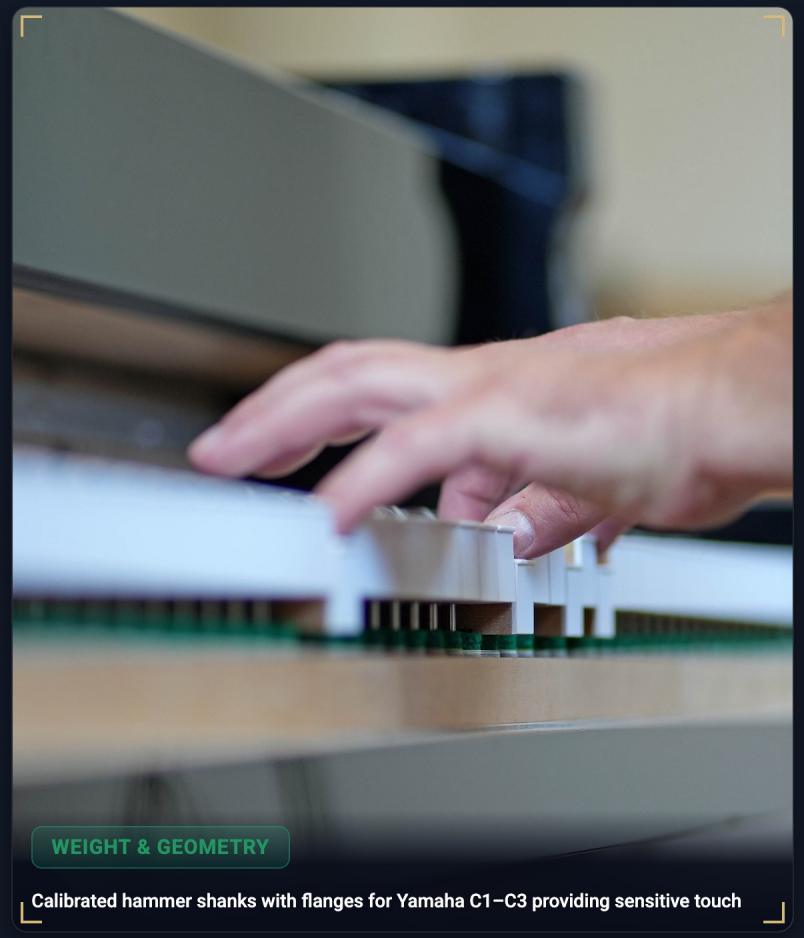
TOUCH

Balanced Weight & Fast Repetition

Calibrated walnut moulding with fine-grained shanks for noticeably lighter key response.

- **Calibrated Walnut Moulding:** Lightweight tonewood moulding reduces inertia for agile, effortless playing.
- **Selected Hornbeam:** Fine-grained quarter-sawn wood guarantees maximum stiffness and pure vibration.
- **Pre-Tensioned Knuckles:** Glued under high tension for superior resilience and instant repetition.

✦ Noticeably Lighter Touch: Reduced inertia for precise dynamic control and rapid repetition

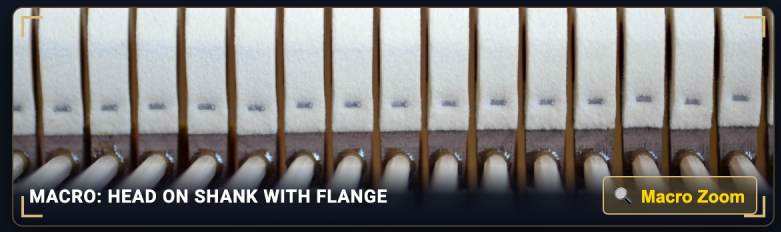


Calibrated hammer shanks with flanges for Yamaha C1–C3 providing sensitive touch

RENNER YAMAHA KIT

Heads & Shanks in Detail

100% drop-in pre-assembled for Yamaha C1, C2, and C3.



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Louis Renner Yamaha Kit (C1–C3) – Seamless Drop-In

Tailored for Yamaha C1–C3: Dynamic expressiveness, brilliant overtones & deep fundamental bass.

SYSTEM — MEASUREMENT SYSTEM | TOUCH — WEIGHT & TOUCH | **DROP-IN — SCREW IN & ALIGN** | TONE — CONCERT TONE | TIME SAVED — HOURS INSTEAD OF DAYS

SCREW IN · ALIGN · DONE

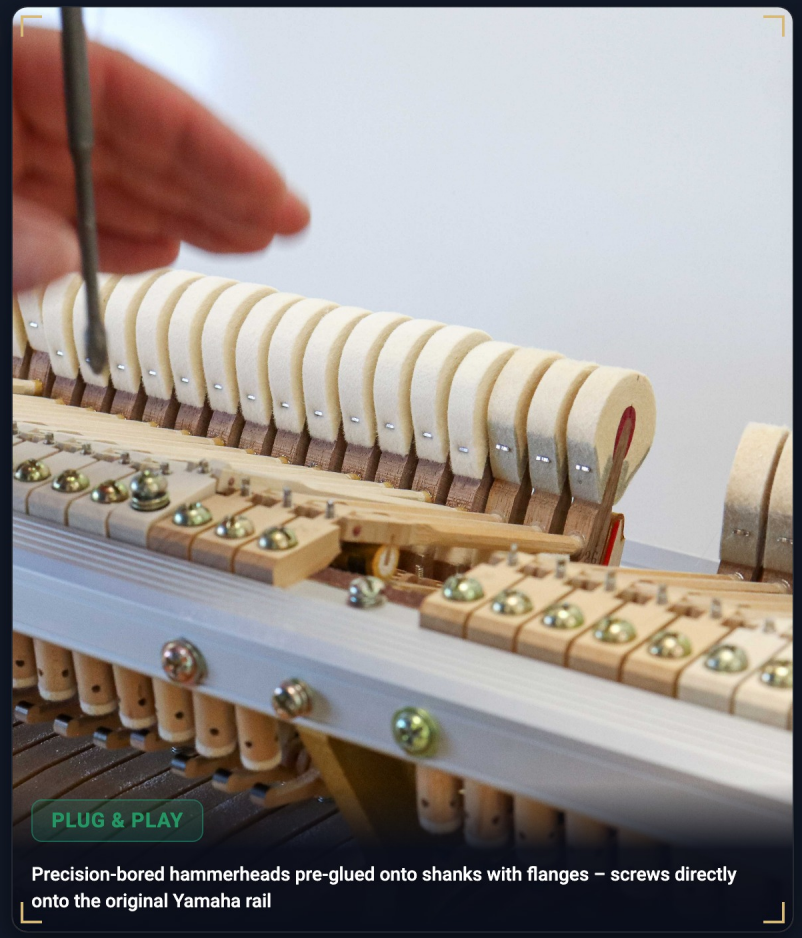
DROP-IN

Simply screw in, align – done.

Factory-bonded onto flanges at the exact angle using traditional natural hide glue.

- **1-Screw Mounting:** Custom flanges screw note by note directly into the original Yamaha rail.
- **Pre-Bored Accuracy:** Pre-drilled at the exact angle – zero measuring or boring in the workshop.
- **Glass-Hard Hide Glue:** Traditional hot hide glue cures rock-hard for lossless vibration transfer into the shank.

✦ Peak Process Reliability: Zero risk of drilling or gluing errors

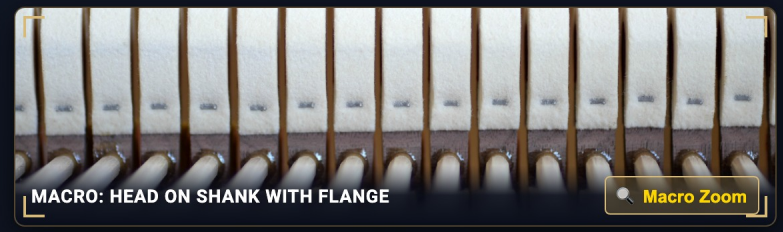


Precision-bored hammerheads pre-glued onto shanks with flanges – screws directly onto the original Yamaha rail

RENNER YAMAHA KIT

Heads & Shanks in Detail

100% drop-in pre-assembled for Yamaha C1, C2, and C3.



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Tailored for Yamaha C1–C3: Dynamic expressiveness, brilliant overtones & deep fundamental bass.

HARMONIC-RICH CONCERT TONE

TONE

Rich Harmonic Spectrum & Color

From warm European pianissimo to powerful projection in fortissimo.

- **Seamless Felt Strip:** Pressed from a single strip — guarantees seamless transitions across all 88 keys.
- **Articulated Bass:** Balanced mass and resilient felt for deep fundamental tone and clean note separation.
- **Bell-Clear Treble:** Lightweight moulding and targeted crown compression for a singing, carrying treble tone.

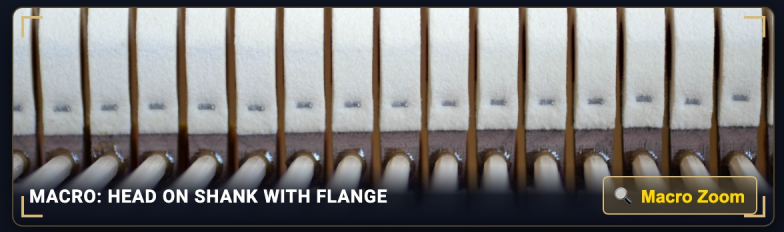
✦ Acoustic Elevation: Transforms harsh attack into warm, singing concert tone

Crafted from a single homogeneous felt strip: Seamless register transitions

RENNER YAMAHA KIT

Heads & Shanks in Detail

100% drop-in pre-assembled for Yamaha C1, C2, and C3.



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HOURS INSTEAD OF DAYS

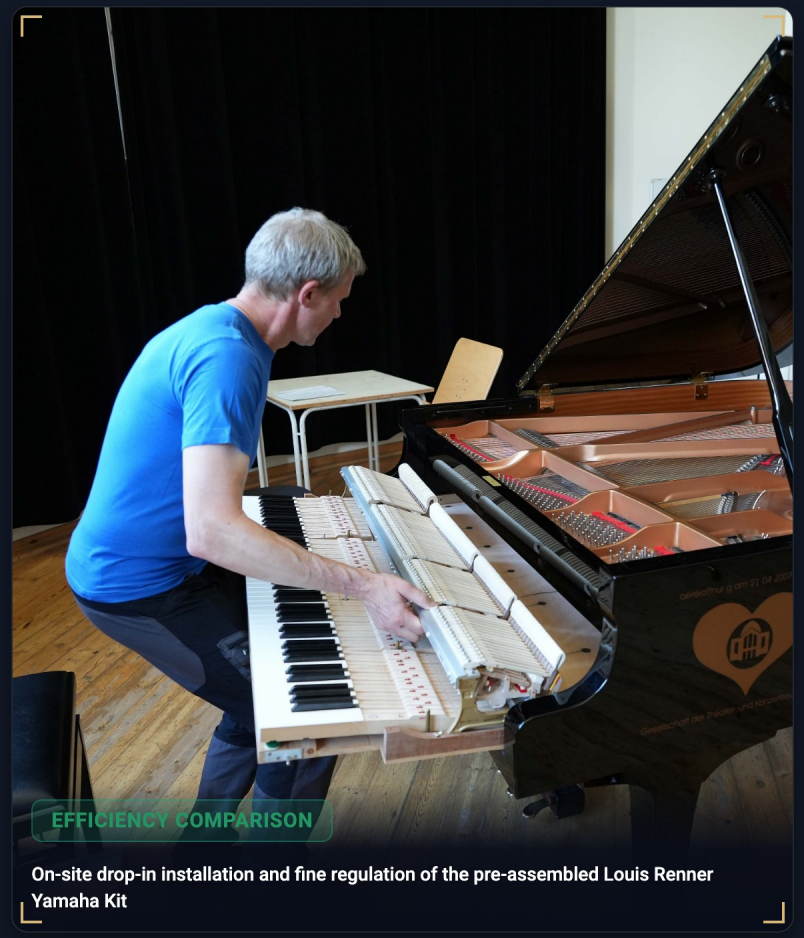
TIME SAVED

4–5 Hours on Site Instead of 3–7 Workshop Days

Traditional multi-day workshop procedure versus same-day installation and regulation on site.

- **Traditional (3–7 Days):** Transport, ungluing, boring chart, hot glue, aligning, sanding & pre-voicing.
- **Renner Kit (4–5 h):** Bolt on directly, align string level and crown – finish voicing if desired.
- **On-Site & Room Acoustics:** Zero transport risk — more time to tailor the sound to room acoustics and customer wishes.

✦ Peak Economic Efficiency: Work reduced from days to a few hours on site

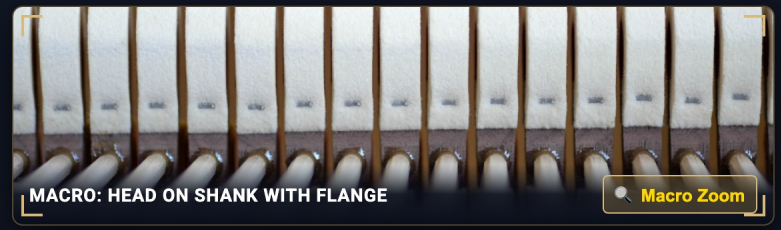


On-site drop-in installation and fine regulation of the pre-assembled Louis Renner Yamaha Kit

RENNER YAMAHA KIT

Heads & Shanks in Detail

100% drop-in pre-assembled for Yamaha C1, C2, and C3.



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PART 4: YAMAHA KIT • FIELD PROOF INDIA (1/2)

Time-Lapse Installation: Complete Conversion in One Day

Documentation of the same-day installation (10:00 AM – 6:00 PM) on site in India (10/2026)

1 Workday

TIMELAPSE · 10–18 H

Complete Action Upgrade Directly On Site

Time-lapse from India (10:00–18:00): Complete on-site conversion including removal, mounting, regulation, tuning & test play.

- **Same-Day Conversion:** Complete upgrade to the pre-assembled Renner Kit from 10:00 to 18:00 on site.
- **Plug & Play Fit:** Direct mounting of pre-calibrated flanges and shanks onto the original rail.
- **On-Site Regulation:** Precise alignment, blow distance, let-off, and checking set right at the grand.
- **Zero Transport Risk:** The grand stays in place and is voiced directly to the room's acoustics.

04

⏱ 10:00–18:00: From removal to a concert-ready grand on site

PART 4: YAMAHA KIT • FIELD PROOF INDIA (2/2)

Pianist Feedback: Pianist Raaga in Action

Immediate artist evaluation of tone color, dynamic range, and repetition on the converted Yamaha grand

100% Concert

ARTIST SOUND PROOF

Immediate Artist Verdict After Installation

Pianist Raaga tests the upgraded Yamaha grand first-hand for tonal warmth, projection, and touch.

- **Singing Concert Tone:** Audible shift from a hard attack to a warm, rich, projecting sound.
- **Blooming Treble:** Bell-like melody projection paired with clean chord clarity in the bass.
- **Nuanced Dynamics:** Effortless shading and tonal control from whisper-soft pp to full ff.
- **Effortless Repetition:** Crisper key response and rapid return thanks to calibrated hammer mass.

04

Immediately audible shift to European warmth and touch control

PART 4: YAMAHA KIT • ACOUSTIC ANALYSIS (1/3)

Sound Spectrum Comparison: Original vs. Renner Yamaha Kit

Original Hammerheads (0:00–5:17) vs. Louis Renner Yamaha Kit (5:18–11:42)

REGISTER: A0–A2 27–110 Hz | A1–A3 55–220 Hz | A2–A4 110–440 Hz | A3–A5 220–880 Hz | A4–A6 440–1760 Hz | A5–A7 880–3520 Hz | Full 25–10k Hz

2 Oct. + H1–H8 | 2 Octaves | Full | Log | Lin

Start Note: A2 – A4 (110–440 Hz)

● Original | ● Yamaha Kit | ▲ Gain | ⋮ H1–H8 | ⤒ Peak Level

1.000 Hz · Orig: -38,4 dB · Kit: -35,9 dB · Δ: +2,53 dB (+79 %)

2-OCTAVE METRICS

Metrics in 2-Octave Window

2-OCTAVE WINDOW
A2 – A4
25 semitones · f_0 = 110–440 Hz

OVERTONE GAIN (H2–H8)
+3.19 dB (+108 %)
Peak at H8 (+4.67 dB)

FUNDAMENTAL H1 (F_0)
-38.4 → -41.2 dB
Δ H1: -2.75 dB

OVERTONE ROLL-OFF
+2.88 dB/kHz
Orig: -9.05 → Kit: -6.17 dB/kHz

✦ ACOUSTIC DIAGNOSIS

Middle Register (A2–A4): H2 (+1.6 dB), H3 (+3.2 dB), H4 (+3.5 dB) & H8 (+4.7 dB) gain rich tonal fullness.

💡 Click piano bar or select Tab 02.

PART 4: YAMAHA KIT • ACOUSTIC ANALYSIS (2/3)

Partial Distribution (H_1–H_{12}) & 24-Semitone Comparison

Overtone architecture per 2-octave window: More warmth and brilliance

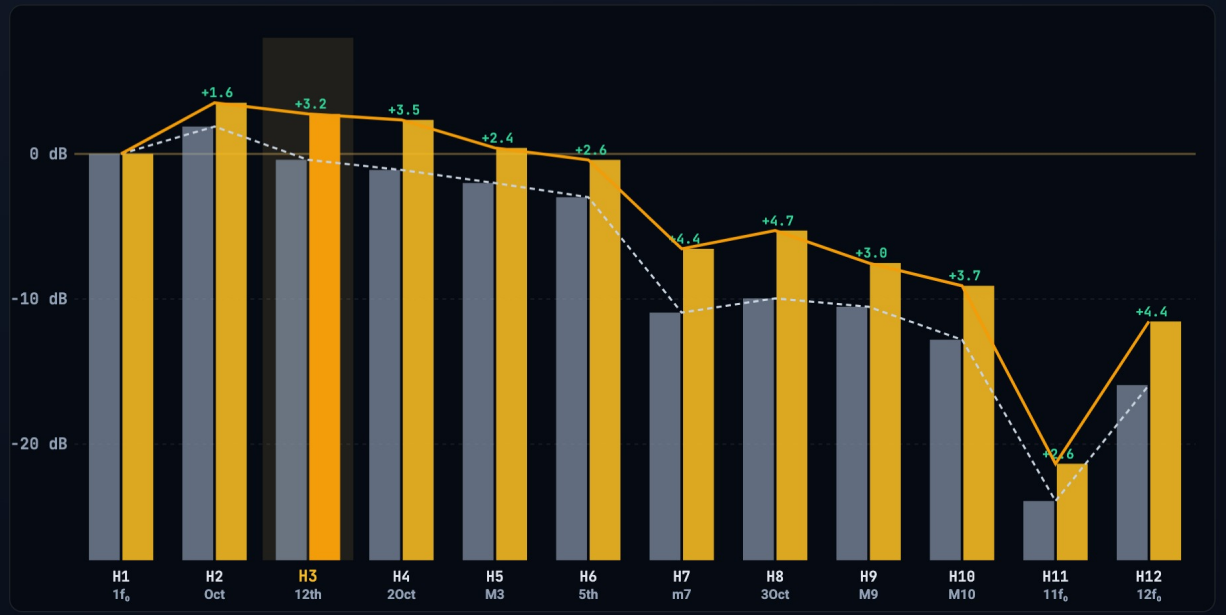
REGISTER: A0–A2 Bass | A1–A3 Tenor | A2–A4 Middle | A3–A5 Treble 1 | A4–A6 Treble 2 | A5–A7 Brilliance | Full A0–C8

Start Note: A2 – A4 (110–440 Hz)

MODE: Relative H_1 | Δ Gain | Absolute (dB)

PARTIAL SERIES H_1–H_{12}

Harmonic Partial Distribution (up to 6 kHz)

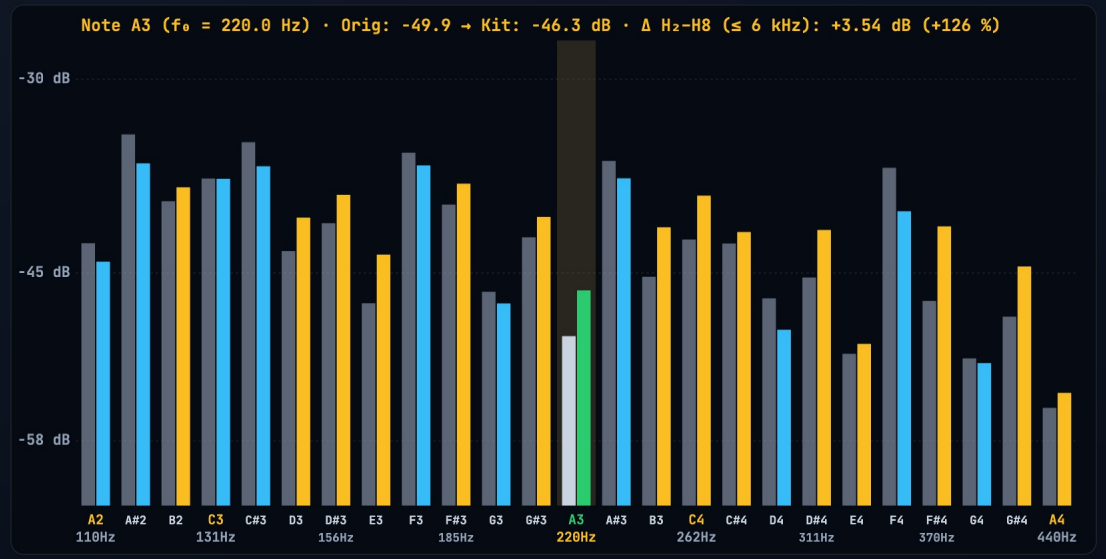


H3 • Twelfth (3·f₀) | A2–A4: 330–1320 Hz (≤ 6 kHz) | Δ: +3.16 dB (+107 %)

ROLL-OFF (H_1→H_8)	EVEN (H_2, H_4, H_6, H_8)	ODD (H_3, H_5, H_7)
+1.32 dB/oct	+3.08 dB	+3.32 dB
-3.5 → -2.2 dB/oct	Warmth & Body	Brilliance & Core

24 CHROMATIC SEMITONES

Note-by-Note Overtone Energy (H_2–H_8)



✦ PARTIAL DIAGNOSIS +3.19 dB (+108 %)

Middle Register (A2–A4): H2 (+1.6 dB), H3 (+3.2 dB), H4 (+3.5 dB) & H8 (+4.7 dB) gain rich tonal fullness.

PART 4: YAMAHA KIT · ACOUSTIC ANALYSIS (3/3)

4 Tonal Gains & Overtone Retention up to C8

Measurable gain in the singing and brilliance registers plus flatter overtone roll-off up to 4,300 Hz

01 · CANTABILE

+2,53 dB (+79 %)

Singing Register (0.8–2.5 kHz)

Up to +3.4 dB at 1.45 kHz for a warm, singing concert tone.

02 · PROJECTION

+1,87 dB (+54 %)

Treble Brilliance (2.5–6.0 kHz)

Carrying overtone brilliance without harshness thanks to Wurzen Weickert felt.

03 · CLARITY

−3,28 dB (−53 %)

Bass Precision (20–250 Hz)

Reduced low-end boominess for clearly separated bass chords and defined fundamentals.

04 · OVERTONE RETENTION

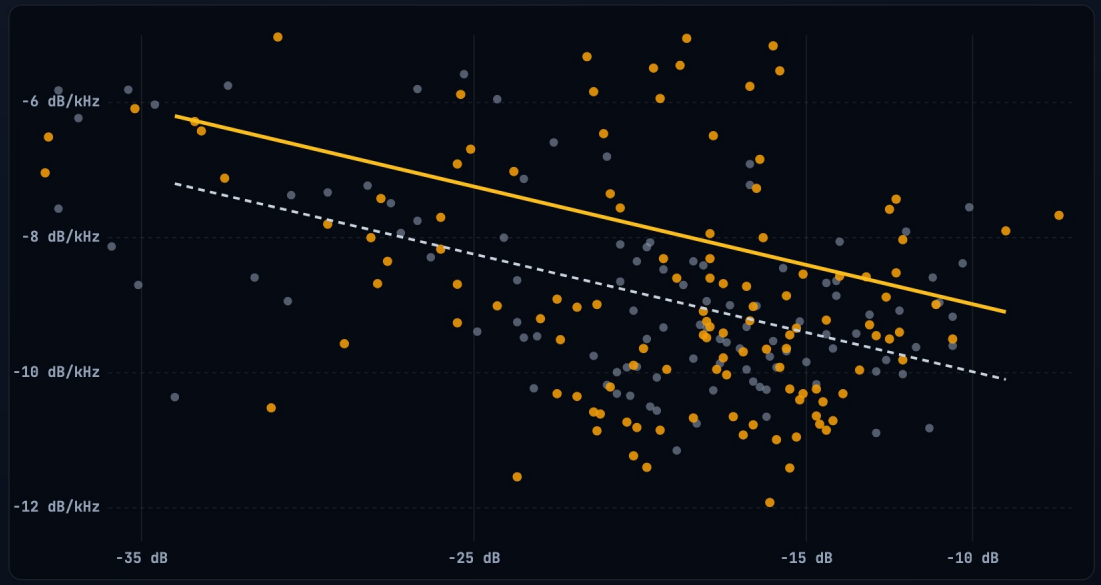
+0,93 dB/kHz (+1,52 dB/Okt.)

12.6% Flatter Roll-Off & Crisper Attack

Stronger upper partials (−6.45 vs. −7.38 dB/kHz) and +0.42 dB attack crest.

OVERTONE ROLL-OFF (TO C_8 = 4,300 HZ)

Overtone Roll-Off by Touch (pp–ff) & Frequency Calculator



PITCH GAIN CALCULATOR (UP TO C_8 = 4,300 HZ):

A4 (440 Hz) | C6 (1047 Hz) | D#7 (2500 Hz) | A7 (3520 Hz) | C8 (4186 Hz)

Frequency f: 2500 Hz (2.50 kHz · Dis7)

ORIGINAL (−7.38 DB/KHZ)	YAMAHA KIT (−6.45 DB/KHZ)	NET GAIN
-18.45 dB	-16.13 dB	+2.32 dB (+71 %)

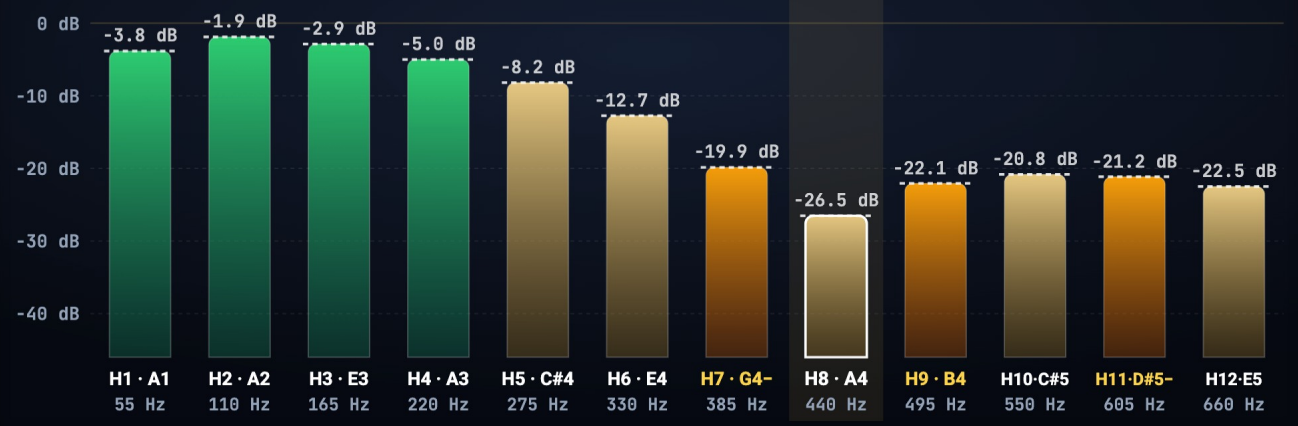
PART 5: VOICING • OPENING: PIANO TONE PHYSICS & PARTIALS

Partial Structure on A1 (55 Hz): Felt Hardness & Strike Point

How felt elasticity and strike ratio (x_0 / L) shape partials H_1–H_{12} in decibels (dB)

Soft | ✦ Concert Ideal | Hard | Level + ΔdB | ΔdB Shift | Play A_1 (55 Hz)

--- Concert Ideal | Green: H_1–H_4 | Gold: Consonant | Amber: Dissonant (H_7·H_9·H_{11})



H8 · A4 (440 Hz · 3rd Oct) | Ref -26.5 dB → -26.5 dB | Δ ±0.0 dB (Felt ±0.0 dB | x_0 ±0.0 dB)

STRIKE POINT X_0 / L — **Standing Wave A_1 (1,560 mm) & Node Cancellation** — H8 (8 antinodes) · Click bar H_1–H_{12}

1. Tonal Hardness (Felt) — Concert Ideal · τc ≈ 2.4 ms

← Softer (needled) | 0% Reference | Harder (compressed) →

2. Strike Point (x_0 / L) — 1/8.0 L (12.5 % · 195 mm)

1/12 L (8.3%) | 1/9 L (H_9) | **1/8 L (Std.)** | 1/7 L (H_7) | 1/5 L (H_5)

FOUNDATION (H_1–H_4)	OVERTONES (H_6–H_{12})	NODE SUPPRESSION
±0.0 dB	±0.0 dB	H8 (-15.7 dB)
55–220 Hz Core & Warmth	330–660 Hz Brilliance	A4 (440 Hz) suppressed

✦ TONAL EFFECT

Concert Balance: Rich Core & Controlled Crown

Balanced elasticity (**τc ≈ 2.4 ms**) blends full bass foundation (H_1–H_4) with singing H_5 and H_6.

x_0 = 1/8 L (195 mm): Node on **H_8 (−15.7 dB)**, tames dissonant H_7 & H_9.

PART 5: VOICING • SPRING PHYSICS, COMPACTION ZONES & DYNAMIC LIMITS

The Hammerhead as a Complex Spring: From Cristofori to Zone Balance

How elasticity, 2:1 compaction (crown → ½ flank), and shoulder transition govern playable dynamics (pp–ff)

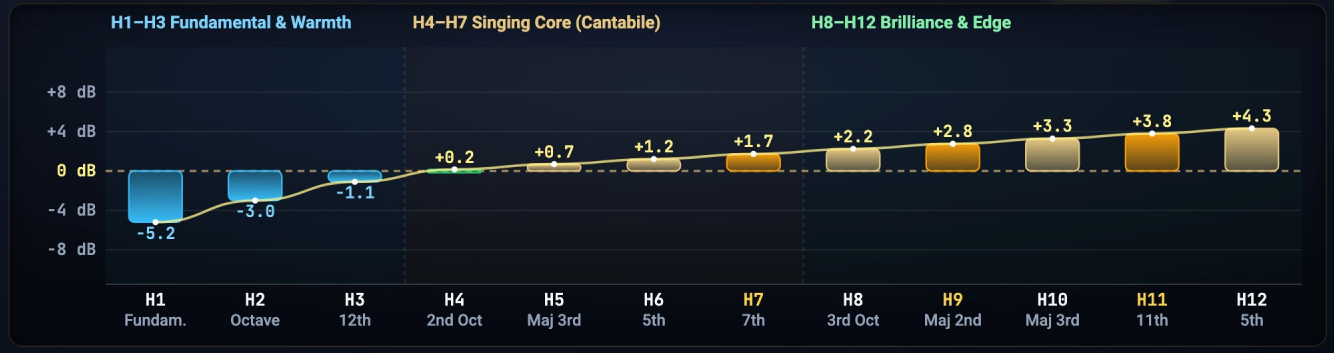
1. Cristofori Roll (1700) | 2. Renner Tension Spring | Ideal

Hollow parchment ring + leather — Silvery-clear (p–f) (+2.4 dB)

PARTIAL STRUCTURE

Change in Partial Structure (H1–H12 in dB)

ΔdB Shift | Level + ΔdB



1. Felt Elasticity & Tension			95 %
1 · Sluggish	2 · Medium	**3 · Elastic**	
2. Hammerhead Shape (Radius)			R ≈ 12.6 mm
1 · Wide	**2 · Concert**	3 · Pointed	
3. Crown Compaction (Flank = ½)			Compaction 50%
1 · Soft	**2 · Ideal**	3 · Too Hard	
4. Crown → Flank Transition			Open (100% Flank)
1 · Open	2 · Firm	3 · Locked ✕	

Playable Dynamics (Outcome) — ✓ Full Range (pp – ff)

pp	mp	mf	f	ff

✓ pp open — ✓ ff open (short τc)

RESTORING FORCE	CONTACT TIME TC	TONAL CHARACTER
95 %	1.62 ms	+2.4 dB
5% hysteresis loss	Crisp ring rebound	Silvery-clear (p–f)

✦ CRISTOFORI (1700)

Leathered Roll: Tension Creates Elasticity

Deer leather shapes gentle *piano*; ring tension (**95% bounce**, **τc ≈ 1.62 ms**) releases the string crisply.

Core Insight: Tension keeps τc short — dead stiffness lingers and damps the string.

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The Hammerhead as a Complex Spring: From Cristofori to Zone Balance

How elasticity, 2:1 compaction (crown → ½ flank), and shoulder transition govern playable dynamics (pp–ff)

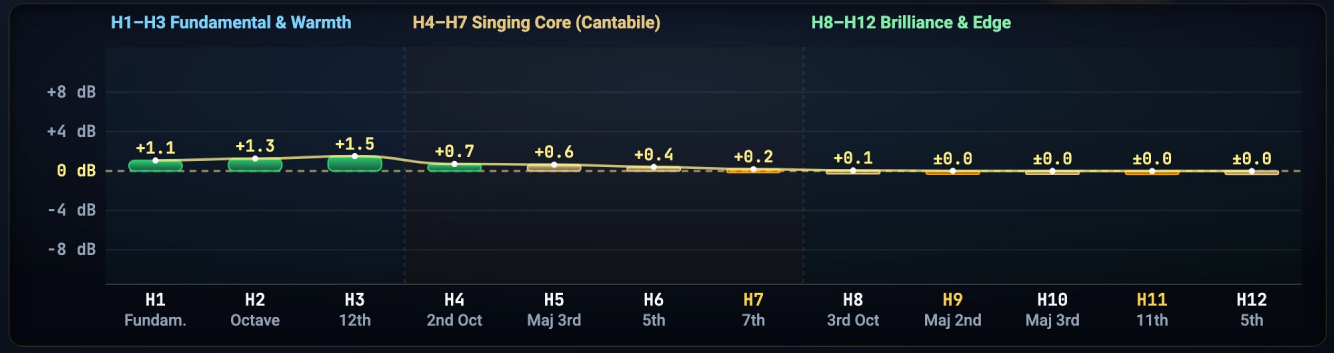
1. Cristofori Roll (1700) | 2. Renner Tension Spring | Ideal

Crown 80% → Flank 40% (½) — Singing (pp–ff) (+0.7 dB)

PARTIAL STRUCTURE

Change in Partial Structure (H1–H12 in dB)

ΔdB Shift | Level + ΔdB



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1 · Sluggish / 2 · Medium / 3 · Elastic	
2. Hammerhead Shape (Radius)	R ≈ 12.6 mm
1 · Wide / 2 · Concert / 3 · Pointed	
3. Crown Compaction (Flank = ½)	Crown 80% · Flank 40%
1 · Soft / 2 · Ideal / 3 · Too Hard	
4. Crown → Flank Transition	Open (100% Flank)
1 · Open / 2 · Firm / 3 · Locked ✕	

Playable Dynamics (Outcome) — ✓ Full Range (pp – ff)

pp | mp | mf | f | ff

✓ pp open — ✓ ff open (short τc)

RESTORING FORCE	CONTACT TIME TC	TONAL CHARACTER
94 %	2.13 ms	+0.7 dB
6% hysteresis loss	Short τc · free ff	Singing (pp–ff)

✦ ZONE BALANCE (PP ↔ FF)

Supple Transition for pp — Elastic ½-Flank for ff

Force flows smoothly from **Crown (80%)** into the **½-Flank (40%)**: delicate *pp* and short **τc ≈ 2.13 ms** for full *ff* (**+0.7 dB**).

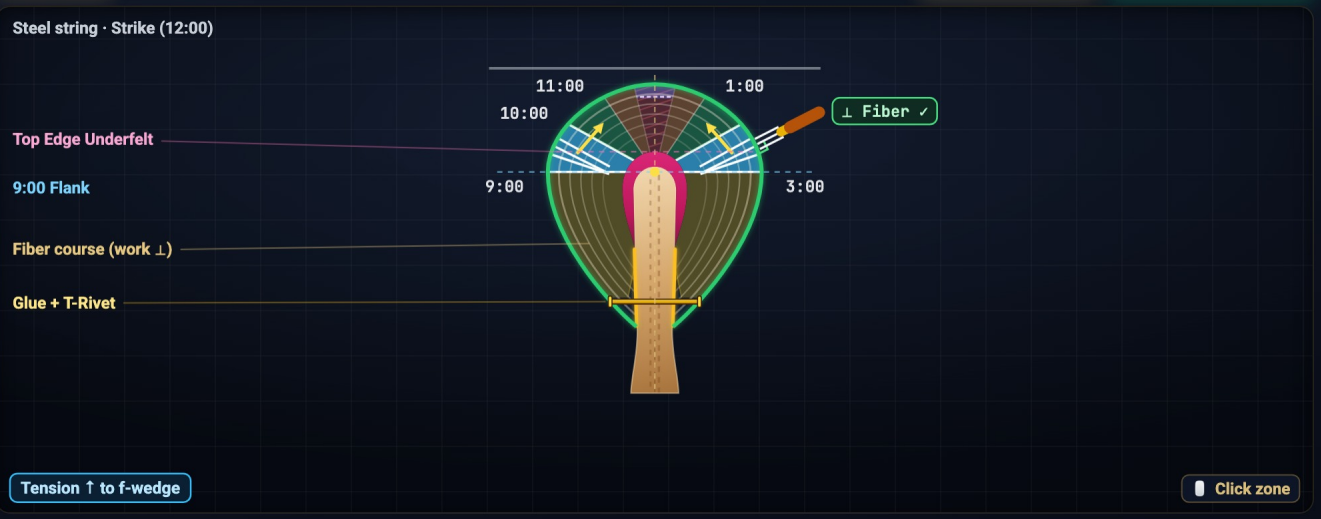
Key Rule: Full dynamics emerge only when gentle onset and fast flank rebound work together.

PART 5: VOICING • FIBER ALIGNMENT, NEEDLING ZONES & TONAL EFFECT

Voicing Zones & Needling Effect: The Renner Clock Model

Always perpendicular to the wool fiber (⊥): from the 9:00–3:00 wedges to the elasticity stitch above the rivet

Blue (ff) | Green (f) | Orange (p) | Purple (pp) | Rivet→Flank | 12:00 Taboo | Needle Pulse | Needle & Play



Blue ff-Wedge (9–10 / 2–3h) | 7–9 mm (pre-UF) · ⊥ fiber | H1–H4 +3.2 dB · open ff

Alignment to Wool Fiber Course

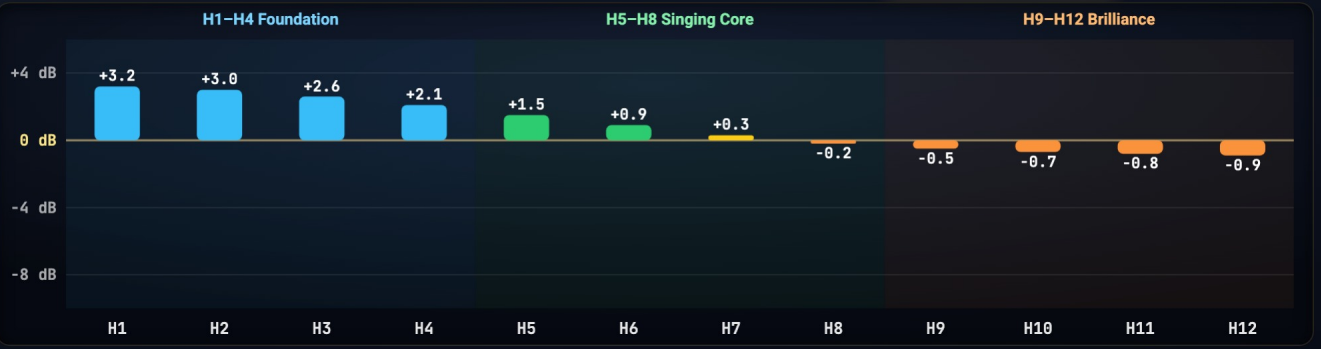
✓ Perp. to fiber · σ intact

1 · Unvoiced | 2 · ✓ ⊥ to Fiber | 3 · ✕ Cross / Chop

- **Blue ff-Wedge (9–10 & 2–3h)** — Deep · ⊥ fiber
  Releases rigidity pre-underfelt, feeds f-wedge ↑ → open ff
- **Green f-Wedge (10–11 & 1–2h)** — 4–6 mm · ⊥ fiber
  Opens transition bridge → singing cantabile core (mf–f)
- **Orange p-Wedge (11–12 & 12–1h)** — 2.5–4 mm · ⊥ fiber
  Cushions bright partials (H8–H12) → warm piano tone
- **Purple pp-Wedge (Crown)** — ≤ 2 mm · grooves
  Between string grooves only (⊥ fiber)! Velvet cushion for pp
- **Above Rivet → Toward Flank** — Few · deep ↑
  Few, deep stitches toward flank → boosts elasticity & high partials ↑
- **Red Taboo Zone (12:00 Core)** — ✕ TABOO (> 2 mm)
  Severs tension arch σ → τc ↑, tone turns hollow & dead!

NEEDLING DEPTH & TOOL	SURFACE TENSION	TONAL EFFECT
7–9 mm	98 % · 1.78 ms	+3.2 dB
3-needle · perp. to fiber	✓ Fibers parted, σ intact	H1–H4 ↑ · free ff

NEEDLING EFFECT — Acoustic Effect of Selected Voicing Zone (H1–… | ΔdB Needling Effect | Dynamics pp–ff



✦ PRE-NEEDLING • BLUE FF-WEDGE (9–10 / 2–3H)

Deep pre-needling releases factory rigidity and opens fortissimo

Pressing **perpendicular to the fiber course** up to the underfelt channels elasticity **upward into the f-wedge**.

Effect: Boosts bass fullness (**H1–H4 +3.2 dB**) while keeping the crown intact.

PART 5: VOICING · GEOMETRY IN THE 3-STRING UNISON CHOIR

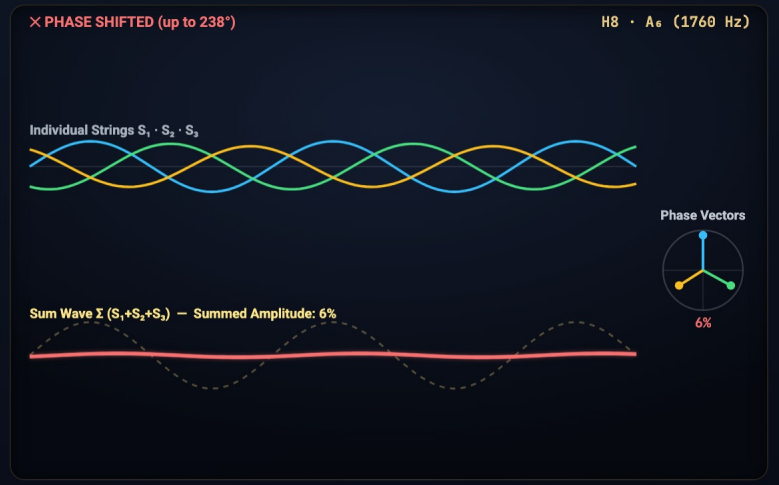
Crown & String Plane in 3-String Choirs: Phase & Cancellation

Even a 0.2 mm height or angle mismatch causes phase cancellation notches across partials H_1–H_{12}

✓ Flat & Parallel | ∠ Tilted Crown | ↕ Uneven String Plane | ⌣ Hollow Crown | Play A_3 Choir

1 · Geometry: Crown ↔ 3-String Plane — Front View

2 · Wave Superposition & Phase Shift — $S_1 + S_2 + S_3 = \Sigma$



PARTIAL SPECTRUM — **Effect on H_1–H_{12} (A_3 · 220 Hz): Phase Cancellation Notches in dB** — Click bar for H_1–H_{12}

H8 · A6 (1760 Hz) · Phase Shift Δφ = 238° — -23.1 dB phase cancellation (6% Σ amplitude)

1. Crown Angle Relative to String Plane — +2.3° · Tilted Crown

← Left lower (−3.5°) | 0.0° Parallel | Right lower (+3.5°) →

2. String Plane / Groove Offset (Δz) — 0.00 mm · Unison perfectly leveled

0.00 mm (Level Plane) | 0.45 mm Offset →

Strike Delay Δt	Peak Cancellation	Coherent Energy
0.37 ms	H8 (-23.1 dB)	63 %
Staggered string contact	Destructive wave interference	Energy loss & false beats

✕ FAULT 1 · TILTED CROWN

Staggered Contact (S_1 → S_2 → S_3) Cancels Upper Partials

A tilted crown strikes S_1 → S_2 → S_3 sequentially. On upper partials, Δφ reaches 120°–240°: wave crests cancel wave troughs!

Effect & Remedy: Metallic buzz & spectral notches. Square hammer travel and file the crown parallel first!

PART 5: VOICING • STRING PREPARATION

String Preparation: Hitch Pins, Bridge Pins & Agraffes

"Even the finest hammerhead can only bring out what is prepared in the instrument!"

4 Prep Steps

STRING BEDDING

Exact String Seating Before the First Needle Stitch

Only firmly seated, level-aligned, and purely tuned string unisons allow an honest tonal diagnosis of the hammerhead.

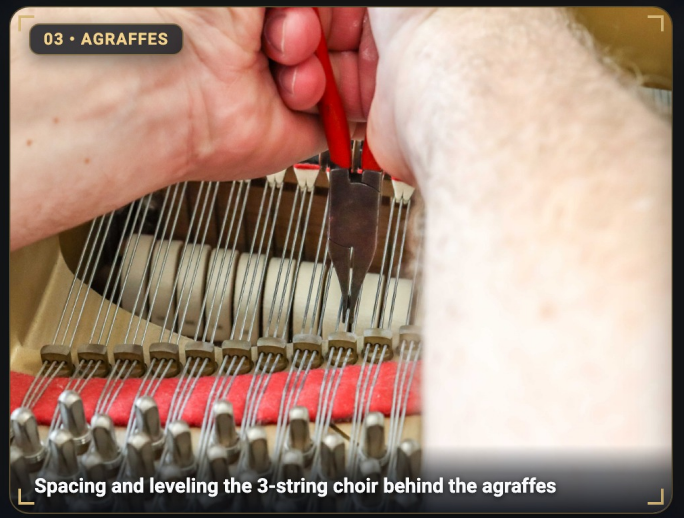
- **1. Hitch Pins & Bearing:** Clean bearing surfaces and check firm, tension-free string seating at the hitch pins.
- **2. Seat at Bridge Pins:** Gently tap strings onto the bridge using a brass drift punch and light hammer to eliminate false beats.
- **3. Agraffes & String Plane:** Level and space the 3 unison strings cleanly behind the agraffes using fine pliers.
- **4. Unison & Damper Check:** Verify simultaneous hammer contact, clean damper seating, and 100% beatless unison tuning.

05

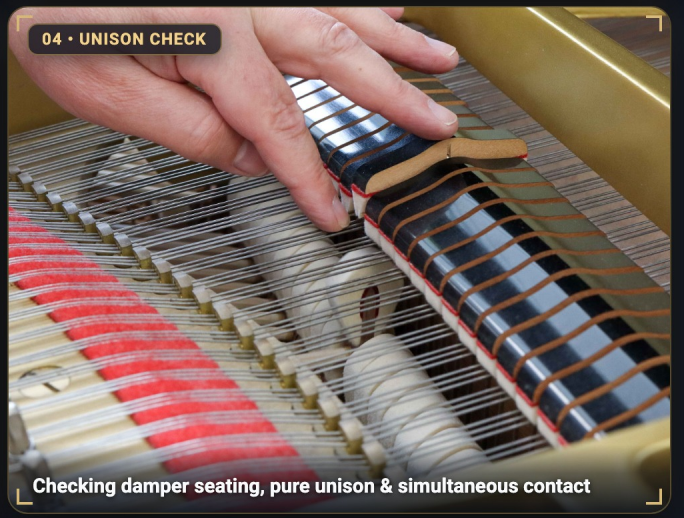
Golden Rule: No needle can fix an unevenly seated or unlevel string unison!

Cleaning bearing surfaces & checking seating at hitch pins

Seating strings cleanly at bridge pins with brass punch



Spacing and leveling the 3-string choir behind the agraffes



Checking damper seating, pure unison & simultaneous contact

PART 5: VOICING • FINE & UNA CORDA VOICING

Intermediate, Fine & Una Corda Voicing

Chromatic equalization, max. 2 mm crown needling & una corda shift-pedal voicing

Max. 2 mm • Una Corda

FINE & SHIFT VOICING

Concert-Grade Equalization—Including the Shift Pedal

All 88 notes blend chromatically—both open and with shift pedal depressed—into a seamless, noise-free tonal progression.

- **1. BDK Fine Holder:** Short single needle in the ergonomic BDK holder (43-0003269) for tactile crown control.
- **2. Mid-Voicing (Shoulder Zone):** Targeted evening-out of outliers in the forte and piano wedges on a firm support pad.
- **3. 2 mm Crown Needling:** With Chop-Stick (43-0003271) between strings – max. 2 mm deep strictly between string grooves!
- **4. Smooth the Crown:** Dress raised wool fibers cleanly with a fine sanding paddle (320–400 grit) for clear attack.
- **5. Shift-Pedal Voicing (Una Corda):** With pedal depressed, voice only contacted string zones & lightly chamfer the left crown edge.

Micro-dosage at the crown & left edge ensures silky shimmer even in una corda!

Short single needle in the BDK holder for delicate crown control

Targeted needling in the piano and forte wedges on support pad

Chop-Stick in the grand: max. 2 mm crown needling strictly between grooves

Fine crown dressing (320–400 grit) & light left-edge chamfer for una corda

PART 5: VOICING • POST-VOICING HARDENING & HAMMER WEIGHT SCALING

Lacquer Dosing, Heat Ironing (max. 135 °C), Crown Finish & Weight

Interactive zone effects of lacquer hardeners, thermal smoothing, micro-polishing (≤ 15 µm) & register-specific hammer mass

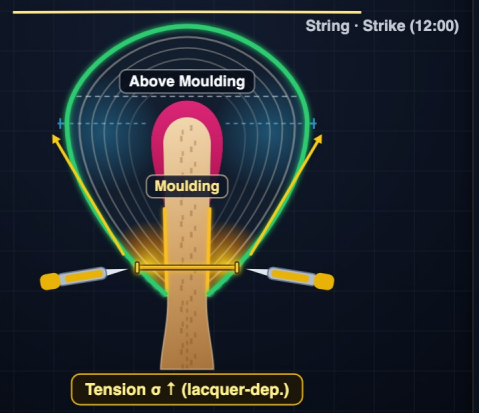
01 • LACQUER DOSING

Lacquer Hardening & Time Compaction

Freshly Applied | Tone Test

Rivet Zone (Foundation) | Lateral Above Moulding | ⚠ Lateral: Too Much | Direct on Crown (Avoid)

✓ Flank free · Spring active

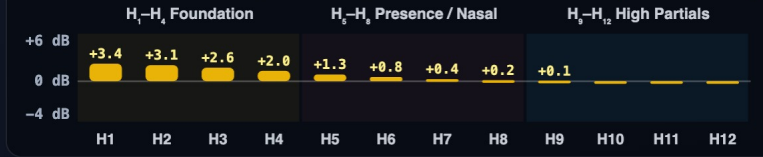


Click any zone on cross-section

Rivet ✓ Spring free · σ ↑ H₁–H₄ +3.4 dB

Partial-Tone Shift (H₁–H₁₂)

Above moulding: com... Spring: 98%



TONAL EFFECT	SPRING & TENSION	OVER TIME
Foundation ↑	Tension ↑	✓ Stable
Reinforces body & bass	Lacquer-dep. · Spring free	Below moulding: stable

✦ FOUNDATION SUPPORT • RIVET AREA

Lacquer at the rivet reinforces the foundation and can lift felt tension

Applied in the rivet area, lacquer stiffens the lower hammer body and—depending on lacquer type—lifts felt tension while leaving flank elasticity untouched.

✓ **Advantage:** Located below the hammer moulding — the elastic impact arch stays protected.

⚖ BOOSTING FUNDAMENTAL TONE & PROJECTION

Control via register-specific hammerhead weight (instead of over-hardening):

+ Middle to Bass: Increase Weight (+m ↑) | − Toward Treble: Reduce Weight (−m ↓)

Middle to Bass → Increase Weight (+m): Higher hammerhead mass drives greater energy into heavy strings, boosting fundamental fullness and punch.

02 • HEAT IRONING

Thermostat-controlled voicing iron smooths and compacts the outer felt

Heat Ironing the Heads (up to max. 135 °C)

Smoothing with a heated iron lays down raised fibers, compacts the outer zone, and lifts shell tension.

- **Max. 135 °C Limit:** Never iron hotter – higher heat damages wool keratin and natural lanolin.
- **Smooth Motion:** Glide evenly along the contour so internal spring elasticity stays intact.

🌡 Thermostat required: Max. 135 °C prevents fiber damage & scorching

03 • ULTRA-FINE CROWN

Micro-abrasive down to 15 µm (left) or Braun Series 1 foil shaver (right)

Micro-Polished Crown (≤ 15 µm) & Shaving

An ultra-smooth crown removes damping fuzz fibers and optimizes string-contact hardness without chemicals.

- **Micro-Abrasive down to 15 µm (15 my):** Polishes the crown mirror-smooth for crisp string contact.
- **Braun Series 1 Foil Shaver:** Standard foil shaver trims protruding fibers with zero felt loss.

✨ Gentlest optimization: Pure tone & 100% preserved spring function

PART 5: VOICING • POST-VOICING HARDENING & HAMMER WEIGHT SCALING

Lacquer Dosing, Heat Ironing (max. 135 °C), Crown Finish & Weight

Interactive zone effects of lacquer hardeners, thermal smoothing, micro-polishing (≤ 15 µm) & register-specific hammer mass

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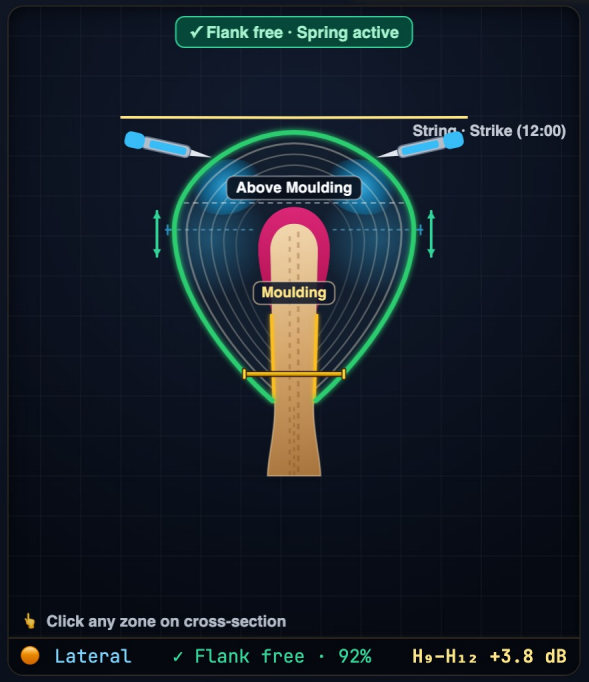

Partial-Tone Shift (H_1–H_{12}) — Above moulding: com... | Spring: 92%

TONAL EFFECT	SPRING & TENSION	OVER TIME
High Partials ↑	✓ Spring Intact	Compacts!
Targeted harmonic boost	Flank link stays open	Above moulding: densifi...

✦ ABOVE HAMMER MOULDING • LATERAL APPLICATION

Lateral dosing above the moulding boosts high partials — and compacts over time!

Lightly applied from the sides above the hammer moulding, diluted lacquer reinforces the upper bridge to boost high partials (H_7–H_{12}) while keeping the outer flank free to flex.

Time Factor: Every application above the hammer moulding compacts further over time during play — always leave a safety margin!

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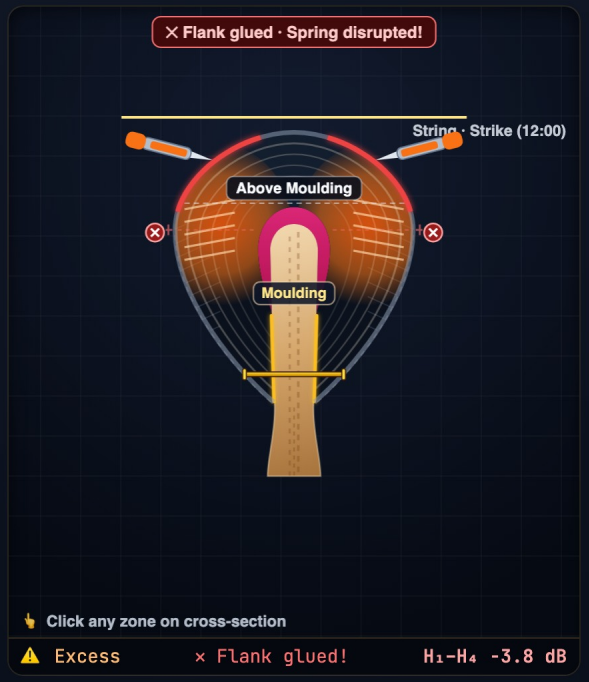
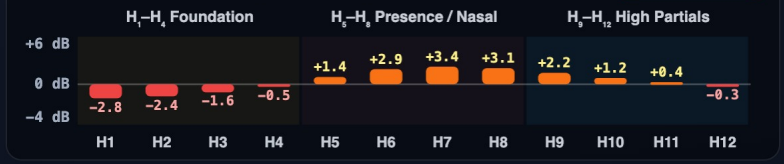


Partial-Tone Shift (H₁–H₁₂) — Above moulding: com... | Spring: 38%



TONAL EFFECT	SPRING & TENSION	OVER TIME
✕ Stiff & Choked	✕ Spring Blocked	⚠ Hardens More
Fundamental drops out	Link to flank glued shut!	Compaction worsens lock

⚠ OVERDOSE ERROR • EXCESS LATERAL LACQUER

Too much lacquer glues the connection to the flank and disrupts spring action!

When over-applied above the hammer moulding, lacquer wicks outward into the flank and glues wool layers together — locking up the hammerhead's complex spring mechanism.

✕ **Result:** Disrupted spring function, lost dynamic range, and progressive hardening over time!

⚖ BOOSTING FUNDAMENTAL TONE & PROJECTION

Control via register-specific hammerhead weight (instead of over-hardening):

+ **Middle to Bass: Increase Weight (+m ↑)** | − **Toward Treble: Reduce Weight (−m ↓)**

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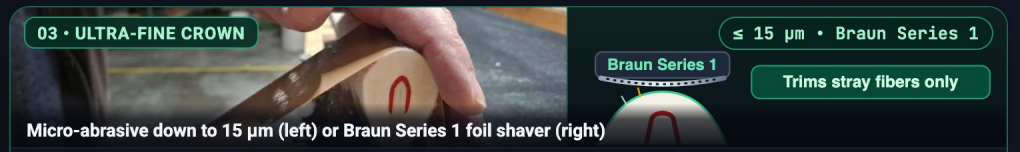
03 • ULTRA-FINE CROWN



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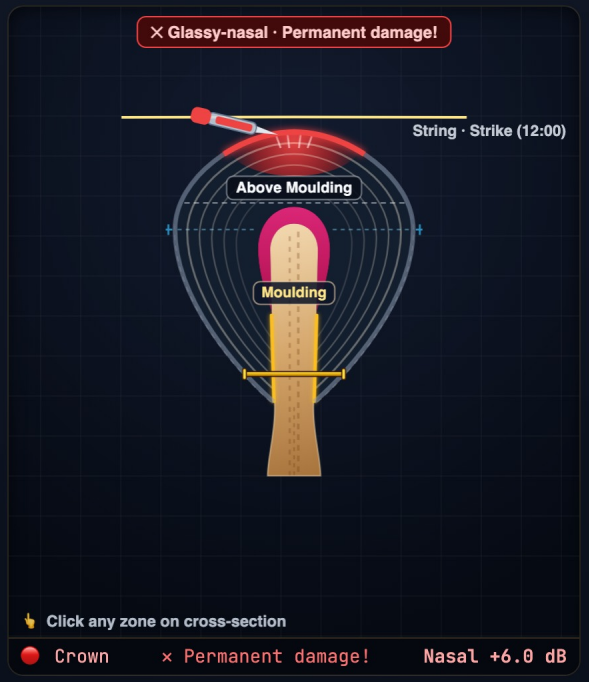

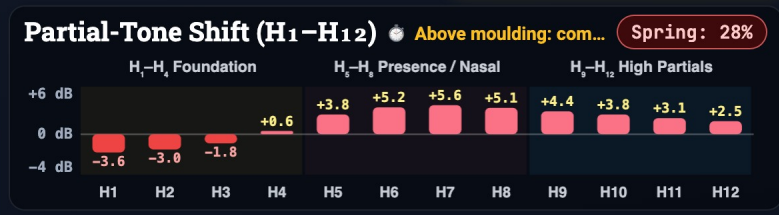

TONAL EFFECT	SPRING & TENSION	OVER TIME
× Glassy-Nasal	× Crown Damaged	× Irreversible
Harsh nasal glare (H_5–H_9)	Micro-cushion destroyed	Compacts harder in play

STRICTLY AVOID • LACQUER DIRECTLY ON CROWN

Lacquer on the crown creates glassy, nasal hardness and permanent damage!

Applying hardener directly to the 12:00 strike crown forms a brittle crust in the string contact zone, destroying surface elasticity while progressive strike compaction worsens the glare.

Workshop Rule: Avoid lacquer on the crown — risk of permanent hammerhead damage!

BOOSTING FUNDAMENTAL TONE & PROJECTION

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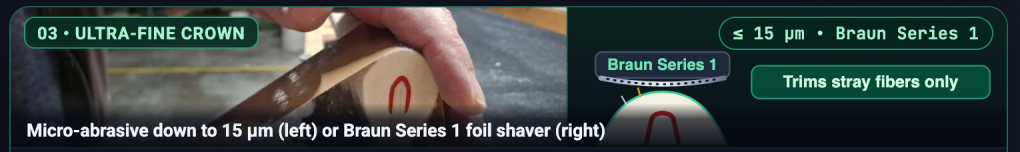

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PART 5: OUTLOOK • RENNER ACADEMY & TECHNICAL DIALOGUE

The Renner Academy & Technical Master Dialogue

Hands-on masterclasses, Meuselwitz factory tours & direct technical partnership worldwide

MASTERCLASSES

Renner Academy Training Program

In-depth workshops covering action geometry, hammerhead installation, Yamaha Kit mounting, and concert voicing.

- **By Masters for Masters:** Hands-on seminars built on authentic Renner factory specifications and gauges.

MEUSELWITZ PLANT

Factory Tours & Direct Technical Support

Experience our unified production in Meuselwitz firsthand and rely on direct workshop support from our master team.

- **Direct Master Dialogue:** Meuselwitz factory visits, custom hammerhead sets & personal technical consultation.

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Your partner for master-grade piano tone since 1882

✦ LOUIS RENNER MASTERCLASS • VIELEN DANK FÜR IHRE AUFMERKSAMKEIT

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Präzision aus Meuselwitz und feinster Wurzener Merinofilz – seit 1882 für den vollkommenen Flügel- und Klavierklang.

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- 📐 Fitting & Yamaha Kit
- 🎵 Voicing & Practice

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